FANUC Robot series

R-30*i*B/R-30*i*B Plus CONTROLLER

AUXILIARY AXIS ORDER MANUAL

For B-cabinet

Robot System Technology Development Department, CNC Hardware Laboratory

Edition	Date	Charge	Check	Approval	Remarks
01	12. Mar. 01	Yuyama	Tanabe	Chino	The first edition publication
02	15. Jun.15	Kino	Enomoto	Tanabe	Add servo hand cable etc.
03	18. Jun.25	Matsumoto	Tanabe	Oya	Add R-30iB Plus etc.

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1. Application

This manual describes auxiliary axis (Aux. Axis) order method for R-30iB/R-30iB Plus B-cabinet.

The scope of this manual as follows.

- If it is necessary the aux. axis by M-2000iA and M-900iA/200P, contact to design section.
- Local stop system is out of scope.
- Gun change function(Hand change function) can be used the (2) axes or (3) axes servo amplifier. However, Brake / Gun change unit cannot be applied to small size Motor.
- In case of M-900iA/600 and M-430iA, servo amplifiers for robot axes are already specified in robot order, so additional servo amplifier has to be added to the robot order.
- The servo amplifier for R-30iA can not be used for spare parts.
- If it is necessary the aux. axis is not supported in this manual, describe the attached Aux. axis configuration sheets and contact to design section.
- If it is necessary the ARP/ARM cable for R-1000iA/120F-7B, contact to design section.

Note.

The power circuit (Breakers, Transformers, Emergency stop unit, Power regeneration unit, Regenerative resistor, etc.) is commonly used with robot and aux. axis.
Therefore, if robot and aux. axis is operated acceleration and de-acceleration with heavy load and high speed simultaneously, the power circuit becomes overloaded, and it may cause overheat of transformer, overheat of regenerative resistor, low voltage of servo amplifier and move error excess.

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2. Available servo motor and servo amplifier

(Note) Use αi-B/βi-B Servo Motor for new order.

2.1. Available αi/βi Servo Motor (DC90V brake)

Table 1.1. αi/βi Servo motor for auxiliary axis of R-30iB/R-30iB Plus Controller.

Servo motor	Specification	Shaft shape	maximum power	Servo amp.
αiF 1/5000	A06B-0202-B605	Taper	2.0kW	20A
	A06B-0202-B705	Straight		
	A06B-0202-B805	Straight with key		
αiF 2/5000	A06B-0205-B605	Taper	2.9kW	20A
	A06B-0205-B705	Straight		
	A06B-0205-B805	Straight with key		
αiF 4/5000	A06B-0223-B605	Taper	4.5kW	40A
	A06B-0223-B705	Straight		
	A06B-0223-B805	Straight with key		
αiF 8/3000	A06B-0227-B605	Taper	5.7kW	40A
	A06B-0227-B705	Straight		
	A06B-0227-B805	Straight with key		
αiF 12/4000	A06B-0243-B605	Taper	7.6kW	80A
	A06B-0243-B705	Straight		
	A06B-0243-B805	Straight with key		
αiF 22/3000	A06B-0247-B605	Taper	9.6kW	80A
	A06B-0247-B705	Straight		
	A06B-0247-B805	Straight with key		
αiS 2/5000	A06B-0212-B605	Taper	2.8kW	20A
	A06B-0212-B705	Straight		
	A06B-0212-B805	Straight with key		
αiS 4/5000	A06B-0215-B605	Taper	3.1kW	20A
	A06B-0215-B705	Straight		
	A06B-0215-B805	Straight with key		
αiS 8/4000	A06B-0235-B605	Taper	8kW	80A
	A06B-0235-B705	Straight		
	A06B-0235-B805	Straight with key		
αiS 12/4000	A06B-0238-B605	Taper	12kW	80A
	A06B-0238-B705	Straight		
	A06B-0238-B805	Straight with key		
αiF 30/4000	A06B-0253-B605	Taper	21kW	160A
	A06B-0253-B705	Straight		
	A06B-0253-B805	Straight with key		
αiF 40/3000	A06B-0257-B605	Taper	18kW	160A
	A06B-0257-B705	Straight		
	A06B-0257-B805	Straight with key		
αiS 22/4000	A06B-0265-B605	Taper	17kW	160A
	A06B-0265-B705	Straight		
	A06B-0265-B805	Straight with key		
αiS 30/4000	A06B-0268-B605	Taper	22kW	160A
	A06B-0268-B705	Straight		
	A06B-0268-B805	Straight with key		
αiS 40/4000	A06B-0272-B605	Taper	24kW	160A
	A06B-0272-B705	Straight		
	A06B-0272-B805	Straight with key		

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Servo motor	Specification	Shaft shape	maximum power	Servo amp.
βiS 0.4/5000	A06B-0114-B704	Straight	0.5kW	20A
	A06B-0114-B804	Straight with key		
βiS 0.5/6000	A06B-0115-B704	Straight	1.3kW	20A
	A06B-0115-B804	Straight with key		
βiS 1/6000	A06B-0116-B704	Straight	2.3kW	20A
	A06B-0116-B804	Straight with key		
βiS 2/4000	A06B-0061-B604	Taper	2.3kW	20A
	A06B-0061-B704	Straight		
	A06B-0061-B804	Straight with key		
βiS 4/4000	A06B-0063-B604	Taper	2.5kW	20A
	A06B-0063-B704	Straight		
	A06B-0063-B804	Straight with key		
βiS 8/3000	A06B-0075-B604	Taper	2.8kW	20A
	A06B-0075-B704	Straight		
	A06B-0075-B804	Straight with key		
βiS 12/3000	A06B-0078-B604	Taper	5.4kW	40A
	A06B-0078-B704	Straight		
	A06B-0078-B804	Straight with key		
βiS 22/2000	A06B-0085-B604	Taper	5.2kW	40A
	A06B-0085-B704	Straight		
	A06B-0085-B804	Straight with key		

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2.2. Available αi-B/βi-B Servo Motor (DC90V brake)

Table 1.2. αi-B/βi-B Servo motor for auxiliary axis of R-30iB/R-30iB Plus Controller.

Servo motor	Specification	Shaft shape	maximum power	Servo amp.
αiF 1/5000-B	A06B-2202-B605	Taper	2.0kW	20A
	A06B-2202-B705	Straight		
	A06B-2202-B805	Straight with key		
αiF 2/5000-B	A06B-2205-B605	Taper	2.9kW	20A
	A06B-2205-B705	Straight		
	A06B-2205-B805	Straight with key		
αiF 4/5000-B	A06B-2223-B605	Taper	4.5kW	40A
	A06B-2223-B705	Straight		
	A06B-2223-B805	Straight with key		
αiF 8/3000-B	A06B-2227-B605	Taper	5.7kW	40A
	A06B-2227-B705	Straight		
	A06B-2227-B805	Straight with key		
αiF 12/4000-B	A06B-2243-B605	Taper	7.6kW	80A
	A06B-2243-B705	Straight		
	A06B-2243-B805	Straight with key		
αiF 22/3000-B	A06B-2247-B605	Taper	9.6kW	80A
	A06B-2247-B705	Straight		
	A06B-2247-B805	Straight with key		
αiS 2/5000-B	A06B-2212-B605	Taper	2.8kW	20A
	A06B-2212-B705	Straight		
	A06B-2212-B805	Straight with key		
αiS 4/5000-B	A06B-2215-B605	Taper	3.1kW	20A
	A06B-2215-B705	Straight		
	A06B-2215-B805	Straight with key		
αiS 8/4000-B	A06B-2235-B605	Taper	8kW	80A
	A06B-2235-B705	Straight		
	A06B-2235-B805	Straight with key		
αiS 12/4000-B	A06B-2238-B605	Taper	12kW	80A
	A06B-2238-B705	Straight		
	A06B-2238-B805	Straight with key		
αiF 30/4000-B	A06B-2253-B605	Taper	21kW	160A
	A06B-2253-B705	Straight		
	A06B-2253-B805	Straight with key		
αiF 40/3000-B	A06B-2257-B605	Taper	18kW	160A
	A06B-2257-B705	Straight		
	A06B-2257-B805	Straight with key		
αiS 22/4000-B	A06B-2265-B605	Taper	17kW	160A
	A06B-2265-B705	Straight		
	A06B-2265-B805	Straight with key		
αiS 30/4000-B	A06B-2268-B605	Taper	22kW	160A
	A06B-2268-B705	Straight		
	A06B-2268-B805	Straight with key		
αiS 40/4000-B	A06B-2272-B605	Taper	24kW	160A
	A06B-2272-B705	Straight		
	A06B-2272-B805	Straight with key		

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Servo motor	Specification	Shaft shape	maximum power	Servo amp.
βiS 0.4/5000-B	A06B-2114-B705	Straight	0.5kW	20A
	A06B-2114-B805	Straight with key		
βiS 0.5/6000-B	A06B-2115-B705	Straight	1.3kW	20A
	A06B-2115-B805	Straight with key		
βiS 1/6000-B	A06B-2116-B705	Straight	2.3kW	20A
	A06B-2116-B805	Straight with key		
βiS 2/4000-B	A06B-2061-B605	Taper	2.3kW	20A
	A06B-2061-B705	Straight		
	A06B-2061-B805	Straight with key		
βiS 4/4000-B	A06B-2063-B605	Taper	2.5kW	20A
	A06B-2063-B705	Straight		
	A06B-2063-B805	Straight with key		
βiS 8/3000-B	A06B-2075-B605	Taper	2.8kW	20A
	A06B-2075-B705	Straight		
	A06B-2075-B805	Straight with key		
βiS 12/3000-B	A06B-2078-B605	Taper	5.4kW	40A
	A06B-2078-B705	Straight		
	A06B-2078-B805	Straight with key		
βiS 22/2000-B	A06B-2085-B605	Taper	5.2kW	40A
	A06B-2085-B805	Straight with key		

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2.3. Available Servo Amplifier

Table 2. Servo amplifiers for auxiliary axis of R-30iB/R-30iB Plus Controller.

Servo Amplifier	Specification (Assembly in Factory)	Specification (Retrofit)	Shape (Width / Fin)	SUMC
αiSV 20	A05B-2605-J040#H103	A06B-6240-H103	60mm / w/o Fin	0.06
αiSV 40	A05B-2605-J040#H104	A06B-6240-H104	60mm / w/ Fin	0.11
αiSV 80	A05B-2605-J040#H105	A06B-6240-H105	60mm / w/ Fin	0.16
αiSV 160	A05B-2605-J040#H106	A06B-6240-H106	60mm / w/ Fin	0.21
αiSV 20/20	A05B-2605-J040#H205	A06B-6240-H205	60mm / w/o Fin	0.11
αiSV 20/40	A05B-2605-J040#H206	A06B-6240-H206	60mm / w/ Fin	0.11
αiSV 40/40	A05B-2605-J040#H207	A06B-6240-H207	60mm / w/ Fin	0.16
αiSV 40/80	A05B-2605-J040#H208	A06B-6240-H208	60mm / w/ Fin	0.21
αiSV 80/80	A05B-2605-J040#H209	A06B-6240-H209	60mm / w/ Fin	0.21
αiSV 80/160	A05B-2605-J040#H210	A06B-6240-H210	90mm / w/ Fin	0.41
αiSV 160/160	A05B-2605-J040#H211	A06B-6240-H211	90mm / w/ Fin	0.41
αiSV 20/20/20	A05B-2605-J040#H305	A06B-6240-H305	60mm / w/o Fin	0.11
αiSV 20/20/40	A05B-2605-J040#H306	A06B-6240-H306	60mm / w/ Fin	0.16
αiSV 40/40/40	A05B-2605-J040#H308	A06B-6240-H308	60mm / w/ Fin	0.17

（Note）Gun change function(Hand change function) can be used the (2) axes or (3) axes servo amplifier. However, Brake / Gun change unit cannot be applied to small size Motor. The servo amplifier for R-30iA can not be used.

Table 3. Applicable servo amplifier for each motors

Model	αiF 1/5000 αiF 2/5000 αiS 2/5000 αiS 4/5000 βiS 0.4/5000 βiS 0.5/6000 βiS 1/6000 βiS 2/4000 βiS 4/4000 βiS 8/3000	αiF 4/5000 αiF 8/3000 βiS 12/3000 βiS 22/2000	αiF 12/4000 αiF 22/3000 αiS 8/4000 αiS 12/4000	αiF 30/4000 αiF 40/3000 αiS 22/4000 αiS 30/4000 αiS 40/4000	SUMA (A)
αiSV 20	○				0
αiSV 40		○			40
αiSV 80			○		80
αiSV 160				○	160
αiSV 20/20	L/M				0
αiSV 20/40	L	M			40
αiSV 40/40		L/M			80
αiSV 40/80		L	M		120
αiSV 80/80			L/M		160
αiSV 80/160			L	M	240
αiSV 160/160				L/M	320
αiSV 20/20/20	L/M/N				0
αiSV 20/20/40	L/M	N			40
αiSV 40/40/40		L/M/N			120

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2.4. Limitation of aux. axis

2.4.1. Limitation of servo amplifier

Four servo amplifiers can be mounted in B-cabinet.
SUMC parameter based on the following calculation should be less than maximum value. If SUMC is exceeded the maximum value, contact to the design section.

Maximum Value of SUMC
In case of Resistor Discharge
1. In case of transformer: 1.00
2. In case of AC reactor: 0.613
In case of Power Regeneration: 1.24
Calculation
Maximum SUMC ＞ Σ SUMC = SUMC(6 axes) ＋ SUMC(AMP2) + SUMC(AMP3-5)

The SUMC of servo amplifier is shown in Table 2.
The SUMC of 6 axes servo amplifier is as follows.

- A05B-2603-H050: 0.41
- A05B-2603-H051: 0.30
- A05B-2603-H052: 0.30
- A05B-2603-H053: 0.20
- A05B-2603-H054: 0.20

2.3.2 The maximum power of aux. axis motors

SUMP is calculated based on the maximum power of each motor.
SUMP is less than the maximum power corresponding to the each transformer capacity.
In case of lack of transformer capacity, it is necessary to upgrade the transformer capacity.
Refer to 11 section regarding the upgrade the transformer capacity.
Refer to 11.6
subsection “Add Aux. Axis of the controller with AC reactor” regarding the upgrade the controller with AC reactor.

Calculation
Maximum SUMP(kW) > Σ maximum power of servo motor x 0.4 to 0.6 (Coefficient)

Maximum power of servo motor is shown in the Table 1.
Coefficient is specified based on the application between from 0.4 to 0.6.
For example, coefficient of rail axis is 0.6, coefficient of Positioner and servo gun is 0.4.
(Note) If SUMP is exceeded the maximum value, Aux axis can be operated by limiting the

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operation of aux. axis (e.g. reduction of acceleration, speed and mass).

Maximum SUMP：

- In case of 3kVA Transformer: 5.3kW or less
- In case of 7.5kVA Transformer: 10.6kW or less
- In case of 10.5kVA Transformer: 14.1kW or less
- In case of 13kVA Transformer: 17.6kW or less
- In case of AC Reactor: 14.1kW or less

2.3.3 Limitation by current of servo amplifier

Total of the current value of the addition axis amplifier must not exceed the maximum SUMA. It is necessary for upgrading the capacity of the transformer from the standard in total of the current value of the addition axis amplifier. The total of the official current of amplifier value in the transformer of each capacity as the addition axis is as follows.
Current value of servo amplifier is shown in the Table 3. If SUMA is exceeded 640A, contact to the design section.

Calculation
Maximum SUMA(A) > Σ Current value of servo amplifier (In case of 20A amplifier, current value is 0(A).)

- In case of 3kVA Transformer: Maximum SUMA = 80A or less
- In case of 7.5kVA Transformer: Maximum SUMA = 160A or less
- In case of 10.5kVA Transformer :Maximum SUMA = 400A or less
- In case of 13kVA Transformer : Maximum SUMA = 640A or less
- In case of AC Reactor : Maximum SUMA = 400A or less

（Note）However, upgrading the capacity of the transformer is unnecessary for FANUC Positioner.

3. Power regeneration option

3.1. αiPS for Power regeneration

In case of Power regeneration, specify PS corresponding to the Robot.
In case of Resistor Discharge, PS is unnecessary.

Name	Specification （Assembly in Factory）	Specification （Retrofit）	Shape (Width / Fin)	Robot
αiPS 15	A05B-2605-J030#H015	A06B-6200-H015	90mm / w / Fin	except M-900iA/600
αiPS 37	A05B-2605-J030#H017	A06B-6200-H037	150mm / w / Fin	M-900iA/600

3.2. αiPS control for Power regeneration

In case of Power regeneration, specify PS control.

Name	Specification
αiPS control	A05B-2607-J010

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4. Auxiliary axis setting

4.1. Selection of motion group (Group Axis number)

Specify the relationship hardware axis and motion group for aux. application.

Motion group	Basic axis	Auxiliary axis
G1	J1-J6	E1-E3
G2	J1-J6	E1-E3
G3	J1-J6	E1-E3
G4	J1-J6	E1-E3
G5	J1-J6	E1-E3
G0	Servo torch, etc.	

4.2. Definition of Axis number

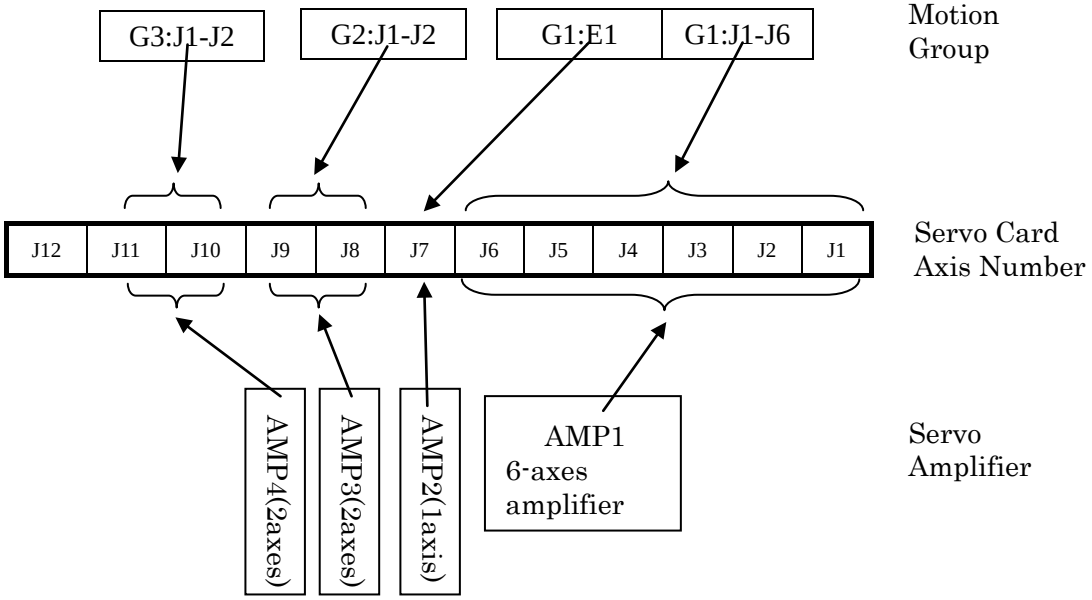
Please define the axis number (J1 – J16) to the motion group.

Define the axis number of servo amplifier and axes.

Axis number of Gun Change/Servo Torch should be last axis.

The motion groups are allocated from smaller Axis Numbers.

Setting example



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4.3. Definition of Brake number

The Brake number(#1.. #16) of each aux. axis have to be assigned when the aux. axis is set up, according to the brake connector(CRR65xx) to be connected.
Refer to 10.1 subsection “Gun change / Brake Unit” about details.

1st controller

Brake number (Software)	#1	#2	#3	#4
6 Axis AMP	CRR65AB	-	-	-
Brake unit	-	CRR65C	CRR65D	CRR65E

2nd controller（Multi arm）

Brake number (Software)	#5	#6	#7	#8
6 Axis AMP	CRR65AB	-	-	-
Brake unit	-	CRR65C	CRR65D	CRR65E

3rd controller（Multi arm）

Brake number (Software)	#9	#10	#11	#12
6 Axis AMP	CRR65AB	-	-	-
Brake unit	-	CRR65C	CRR65D	CRR65E

4th controller（Multi arm）

Brake number (Software)	#13	#14	#15	#16
6 Axis AMP	CRR65AB	-	-	-
Brake unit	-	CRR65C	CRR65D	CRR65E

5. Auxiliary Axis Control/Adaptor plate for Servo amplifier

Specify the adaptor plate and aux. axis control as following tables.
The combination of Servo amplifiers is different in Resistor Discharge and Power regeneration.

5.1. Adaptor plate for servo amplifier

Specify adaptor plate of the same number as “60mm / w/o Fin” Servo amplifier.
Refer to Table 2.

Adaptor plate	Specification
For 60mm servo amplifier w/o Fin	A05B-2605-J420

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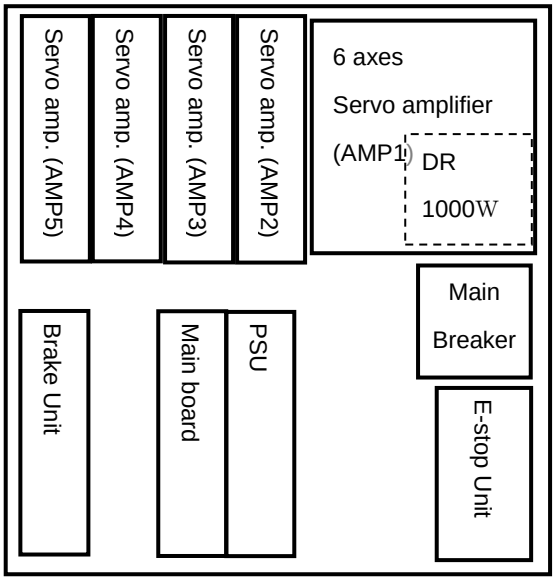
5.2. In case of Regenerative resistor A(1 element/1000W, A05B-2603-H060)

Robot : R-2000iB, R-1000iA, M-710iC, M-420iA, ARC Mate 100iC, ARC Mate 120iC, M-10iA, M-20iA, M-430iA, F-200iB

(Note) The servo amplifiers for Auxiliary axis are mounted up to four.



The combination of Auxiliary axis control and Adaptor plate.

In case of Regenerative resistor A(1 element/1000W, A05B-2603-H060)

AMP5 Width	AMP4 Width	AMP3 Width	AMP2 Width	Auxiliary axis control	Adaptor plate
-	-	-	60mm	A05B-2607-J001	A05B-2607-J430
-	-	-	90mm	A05B-2607-J001	A05B-2607-J431
-	-	60mm	60mm	A05B-2607-J001 A05B-2607-J002	A05B-2607-J432
-	-	90mm	60mm	A05B-2607-J001 A05B-2607-J003	A05B-2607-J433
-	-	60mm	90mm	A05B-2607-J001 A05B-2607-J004	A05B-2607-J434
-	-	90mm	90mm	A05B-2607-J001 A05B-2607-J005	A05B-2607-J435
-	60mm	60mm	60mm	A05B-2607-J001 A05B-2607-J002(X2)	A05B-2607-J436
-	60mm	60mm	90mm	A05B-2607-J001 A05B-2607-J002 A05B-2607-J004	A05B-2607-J455
-	60mm	90mm	90mm	A05B-2607-J001 A05B-2607-J004 A05B-2607-J005	A05B-2607-J456
-	90mm	60mm	90mm	A05B-2607-J001 A05B-2607-J003 A05B-2607-J004	A05B-2607-J457
-	90mm	90mm	90mm	A05B-2607-J001 A05B-2607-J005(X2)	A05B-2607-J458
60mm	60mm	60mm	90mm	A05B-2607-J001 A05B-2607-J002(X2) A05B-2607-J004	A05B-2607-J437

Specify adaptor plate of the same number as “60mm / w/o Fin” Servo amplifier.

<In case of M-430iA series>

Only AMP3 is can be added for aux. axis, because servo amplifier for the robot has already been mounted on AMP2.

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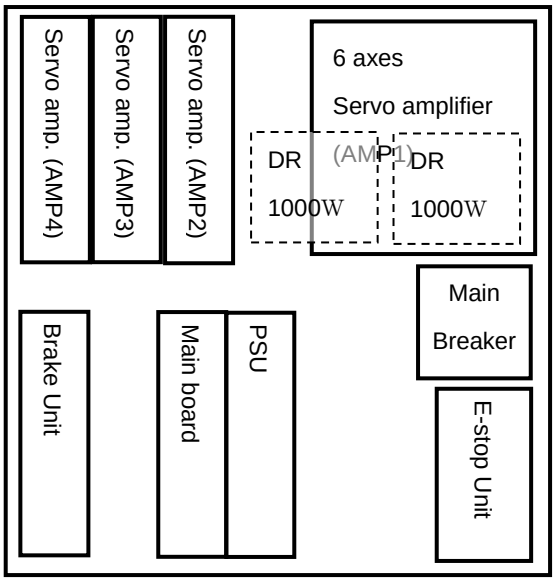
5.3. In case of Regenerative resistor B(2 elements/2000W, A05B-2603-H062)

Robot : R-2000iB/100P, M-410iB, R-2000iB/200T, R-2000iB/220U, M-900iA

(Note) The servo amplifiers for Auxiliary axis are mounted up to three.



The combination of Auxiliary axis control and Adaptor plate.

In case of Regenerative resistor B(2 elements/2000W, A05B-2603-H062)

AMP4 Width	AMP3 Width	AMP2 Width	Auxiliary axis control	Adaptor plate
-	-	60mm	A05B-2607-J001	A05B-2607-J440
-	-	90mm	A05B-2607-J001	A05B-2607-J441
-	60mm	60mm	A05B-2607-J001 A05B-2607-J002	A05B-2607-J442
-	90mm	60mm	A05B-2607-J001 A05B-2607-J003	A05B-2607-J443
-	60mm	90mm	A05B-2607-J001 A05B-2607-J004	A05B-2607-J444
-	90mm	90mm	A05B-2607-J001 A05B-2607-J005	A05B-2607-J445
60mm	60mm	60mm	A05B-2607-J001 A05B-2607-J002(X2)	A05B-2607-J446

Specify adaptor plate of the same number as “60mm / w/o Fin” Servo amplifier.

<In case of M-900iA/600>

Only AMP3 is can be added for aux. axis, because servo amplifier for the robot has already been mounted on AMP2 and AMP3(one axis)

One axis of the remainder of two axis Servo amplifier of AMP3 can be used for Auxiliary axis (Only αiSV80/80 and αiSV80/160).

Specify all Servo amplifiers for Robot axis when you use the order support system.

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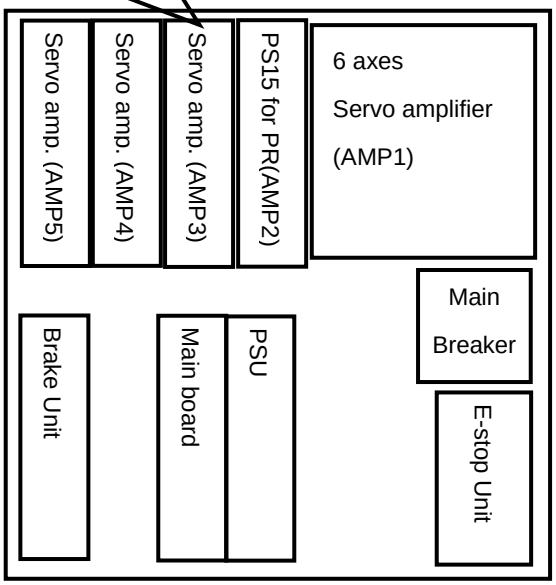
5.4. In case of Power regeneration (except M-900iA/600)

(Note) The servo amplifier number at this position is "AMP3".

(Note) The servo amplifiers for Auxiliary axis are mounted up to three.



The combination of Auxiliary axis control and Adaptor plate.

In case of Power regeneration (except M-900iA/600)

AMP5 Width	AMP4 Width	AMP3 Width	AMP2 αiPS 15	Auxiliary axis control	Adaptor plate
-	-	60mm	90mm	A05B-2607-J011	A05B-2607-J434
-	-	90mm	90mm	A05B-2607-J012	A05B-2607-J435
-	60mm	60mm	90mm	A05B-2607-J002 A05B-2607-J011	A05B-2607-J455
-	60mm	90mm	90mm	A05B-2607-J004 A05B-2607-J012	A05B-2607-J456
-	90mm	60mm	90mm	A05B-2607-J003 A05B-2607-J011	A05B-2607-J457
-	90mm	90mm	90mm	A05B-2607-J005 A05B-2607-J012	A05B-2607-J458
60mm	60mm	60mm	90mm	A05B-2607-J002(X2) A05B-2607-J011	A05B-2607-J437

Specify adaptor plate of the same number as “60mm / w/o Fin” Servo amplifier.

<In case of M-430iA series>

Only AMP3 is can be added for aux. axis, because servo amplifier for the robot has already been mounted on AMP2.

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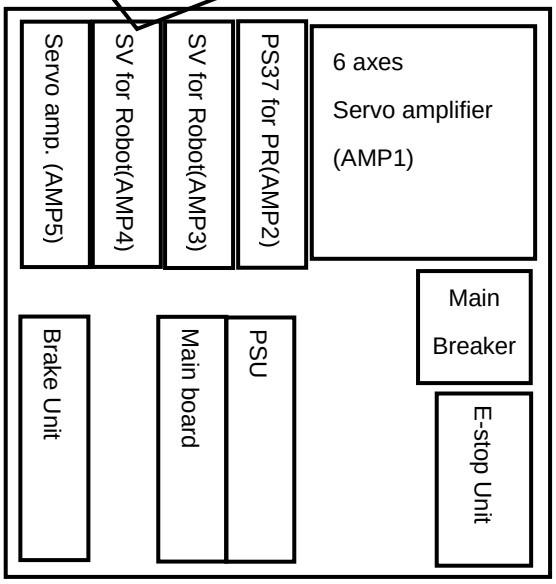
5.5. In case of Power Regeneration (M-900iA/600)

(Note) One axis (80A) of AMP4 is used for Robot.
Use one remainder axis of two axis servo amplifier for Auxiliary axis.

(Note) The servo amplifiers for Auxiliary axis are mounted up to two.



The combination of Auxiliary axis control and Adaptor plate.

In case of Power regeneration (M-900iA/600)

AMP5 Width	AMP4 Width	AMP3 αiSV 80/80	AMP2 αiPS37	Auxiliary axis control	Adaptor plate
-	60mm	60mm	150mm	A05B-2607-J002 A05B-2607-J011	A05B-2607-J450
-	90mm	60mm	150mm	A05B-2607-J003 A05B-2607-J011	A05B-2607-J451
60mm	60mm	60mm	150mm	A05B-2607-J002(X2) A05B-2607-J011	A05B-2607-J452
90mm	60mm	60mm	150mm	A05B-2607-J002 A05B-2607-J003 A05B-2607-J011	A05B-2607-J453
60mm	90mm	60mm	150mm	A05B-2607-J003 A05B-2607-J004 A05B-2607-J011	A05B-2607-J454

Specify adaptor plate of the same number as “60mm / w/o Fin” Servo amplifier.

<In case of M-900iA/600>

Only AMP3 is can be added for aux. axis, because servo amplifier for the robot has already been mounted on AMP2 and AMP3(one axis)

One axis of the remainder of two axis Servo amplifier of AMP3 can be used for Auxiliary axis (Only αiSV80/80 and αiSV80/160).

Specify all Servo amplifiers for Robot axis when you use the order support system.

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6. Servo Card

6 axes servo card is selected in the system without the auxiliary axis.

Change it to the servo card that can control the number of axes in which the number of addition axes is added to the number of robot axes.

Number of control axes (Including robot axes)	Specification	
	R-30iB	R-30iB Plus
6 axes (standard)	A05B-2600-H040	A05B-2670-H040
12 axes	A05B-2600-H041	A05B-2670-H041
18 axes	A05B-2600-H042	A05B-2670-H042

7. Cable Tag

Select tag according to the servo motor and the amplifier composition.

7.1. Brake cable Tag

Brake Number	Connection to	Connector	Specification
1	6 axes amplifier（AMP1）	CRR65AB	A05B-2605-J060
2	Brake unit (BKUNIT)	CRR65C	A05B-2605-J061
	Gun Change Brake unit (GCPCB#1)	CRR65C	A05B-2605-J064
	Gun Change Brake unit (GCPCB#2)	CRR65C	A05B-2605-J065
	Gun Change Brake unit (GCPCB#3)	CRR65C	A05B-2605-J066
3	Brake unit (BKUNIT)	CRR65D	A05B-2605-J062
4	Brake/GC unit (BKUNIT)	CRR65E	A05B-2605-J063

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7.2. Servo motor cable Tag

Position of Amp.	Specification
AMP2/L	A05B-2605-J070
AMP2/M	A05B-2605-J071
AMP2/N	A05B-2605-J072
AMP3/L	A05B-2605-J073
AMP3/M	A05B-2605-J074
AMP3/N	A05B-2605-J075
AMP4/L	A05B-2605-J076
AMP4/M	A05B-2605-J077
AMP4/N	A05B-2605-J078
AMP5/L	A05B-2605-J079
AMP5/M	A05B-2605-J080
AMP5/N	A05B-2605-J081
AMP6/L	A05B-2605-J082
AMP6/M	A05B-2605-J083
AMP6/N	A05B-2605-J084

7.3. Gun change Tag

Specification
A05B-2605-J085

Gun change number	Specification
GCPCB#1	A05B-2605-J085
GCPCB#2	A05B-2605-J086
GCPCB#3	A05B-2605-J087

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8. Auxiliary axis cable

8.1. ARP/ARM cable

Available Servo Motor：αiF1/5000, αiF2/5000, αiF4/5000, αiF8/3000, αiS2/5000, αiS4/5000, αiS8/4000, βiS0.4/5000, βiS0.5/6000, βiS1/6000, βiS2/4000, βiS4/4000, βiS8/3000, βiS12/3000 and FANUC 1 axis positioner

Cable Type : A, Interface : A

Non-Flex / Flex	No. of axis	Standard /NRTL /CE	Length	Specification	Wire	
					Power/Brake (ARM)	Pulsecoder (ARP)
Non-Flex	1	Standard / NRTL	7m	A05B-2607-J100	Diameter: Φ11.8mm Weight: 0.2kg/m	Diameter: Φ8.5mm Weight : 0.09kg/m
			14m	A05B-2607-J101		
			20m	A05B-2607-J102		
			30m	A05B-2607-J103		
Flex			7m	A05B-2607-J105	Diameter: Φ11.8mm Weight: 0.2kg/m	Diameter: Φ9.1mm Weight : 0.12kg/m
			14m	A05B-2607-J106		
			20m	A05B-2607-J107		
			30m	A05B-2607-J108		
Non-Flex		CE	7m	A05B-2607-J110	Diameter: Φ12.5mm Weight: 0.25kg/m	Diameter: Φ8.5mm Weight: 0.09kg/m
			14m	A05B-2607-J111		
			20m	A05B-2607-J112		
			30m	A05B-2607-J113		
Flex			7m	A05B-2607-J115	Diameter: Φ12.5mm Weight: 0.25kg/m	Diameter: Φ9.1mm Weight: 0.12kg/m
			14m	A05B-2607-J116		
			20m	A05B-2607-J117		
			30m	A05B-2607-J118		

8.2. 2Axes ARP/ARM cable

Available Servo Motor：αiF1/5000, αiF2/5000, αiS2/5000, αiS4/5000, βiS0.4/5000, βiS0.5/6000, βiS1/6000

Cable Type : H, Interface : A

Non-Flex / Flex	No. of axis	Standard /NRTL /CE	Length	Specification	Wire	
					Power/Brake (ARM)	Pulsecoder (ARP)
Non-Flex	2	Standard / NRTL	7m	A05B-2607-J120	Diameter: Φ15.2mm(#C) Weight : 0.35kg/m	Diameter: Φ12mm(#C) Weight: 0.23kg/m
			14m	A05B-2607-J121		
			20m	A05B-2607-J122		
			30m	A05B-2607-J123		

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8.3. PM/PP Cable

PM/PP cable is used for FANUC 2 axes Positioner.

Cable Type : B, Interface : B

Non-Flex / Flex	No. of axis	Standard /NRTL /CE	Length	Specification	Wire	
					Power・Brake (PM)	Pulsecoder (PP)
Non-Flex	2	Standard /NRTL	7m	A05B-2607-J160	Diameter: Φ20mm Weight: 0.65kg/m	Diameter: Φ12mm Weight: 0.23kg/m
			14m	A05B-2607-J161		
			20m	A05B-2607-J162		
			30m	A05B-2607-J163		
Flex			7m	A05B-2607-J165	Diameter: Φ20mm Weight: 0.61kg/m	Diameter: Φ20.5mm Weight: 0.71kg/m
			14m	A05B-2607-J166		
			20m	A05B-2607-J167		
			30m	A05B-2607-J168		
Non-Flex		CE	7m	A05B-2607-J170	Diameter: Φ20.6mm Weight: 0.71kg/m	Diameter: Φ12mm Weight: 0.23kg/m
			14m	A05B-2607-J171		
			20m	A05B-2607-J172		
			30m	A05B-2607-J173		
Flex			7m	A05B-2607-J175	Diameter: Φ20mm Weight: 0.64kg/m	Diameter: Φ20.5mm Weight: 0.71kg/m
			14m	A05B-2607-J176		
			20m	A05B-2607-J177		
			30m	A05B-2607-J178		

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8.4. J7P/J7M Cable

J7P/J7M cable is used for Rail axis or FANUC 1 axis positioner

Available Servo Motor: αiF12/4000, αiF22/3000, βiS22/3000 and FANUC 1 axis positioner

Cable Type : , Interface : C

Non-Flex /Flex	No. of axis	Standard /NRTL /CE	Length	Specification	Wire	
					Power・Brake (J7M)	Pulsecoder (J7P)
Non-Flex	1	Standard /NRTL	7m	A05B-2607-J180	Diameter: Φ15.8mm Weight: 0.42kg/m	Diameter: Φ8.5mm Weight: 0.09kg/m
			14m	A05B-2607-J181		
			20m	A05B-2607-J182		
			30m	A05B-2607-J183		
		CE	7m	A05B-2607-J190	Diameter: Φ20.6mm Weight: 0.71kg/m	Diameter: Φ8.5mm Weight: 0.09kg/m
			14m	A05B-2607-J191		
			20m	A05B-2607-J192		
			30m	A05B-2607-J193		

Available Servo Motor: αiF30/4000, αiF40/3000, αiS22/4000, αiS30/4000, αiS40/4000

Cable Type : A, Interface : C

Non-Flex / Flex	No. of axis	Standard /NRTL /CE	Length	Specification	Wire	
					Power・Brake (J7M)	Pulsecoder (J7P)
Non-Flex	1	Standard /NRTL	7m	A05B-2607-J185	Diameter: Φ20mm Weight: 0.65kg/m	Diameter: Φ8.5mm Weight: 0.09kg/m
			14m	A05B-2607-J186		
			20m	A05B-2607-J187		
			30m	A05B-2607-J188		
		CE	7m	A05B-2607-J195	Diameter: Φ20.6mm Weight: 0.71kg/m	Diameter: Φ8.5mm Weight: 0.09kg/m
			14m	A05B-2607-J196		
			20m	A05B-2607-J197		
			30m	A05B-2607-J198		

8.5. Aux. axis Motor connection cable (Directly connected cable)

Available Servo Motor: βiS0.4/5000, βiS0.5/6000, βiS1/6000 / Straight Connector

Cable Type : C, Connector Direction : A

Non-Flex /Flex	No. of axis	Standard /NRTL /CE	Length	Specification	Wire		
					Power	Brake	Pulsecoder
Flex	1	Standard /NRTL	7m	A05B-2607-J300	Diameter: Φ6.4mm Weight: 0.05kg/m	Diameter: Φ6.4mm Weight: 0.05kg/m	Diameter: Φ7mm Weight: 0.09kg/m
			14m	A05B-2607-J301			
			20m	A05B-2607-J302			
			30m	A05B-2607-J303			
		CE	7m	A05B-2607-J305	Diameter: Φ6.4mm Weight: 0.07kg/m	Diameter: Φ6.4mm Weight: 0.07kg/m	Diameter: Φ7mm Weight: 0.09kg/m
			14m	A05B-2607-J306			
			20m	A05B-2607-J307			
			30m	A05B-2607-J308			

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Available Servo Motor: βiS0.5/6000, βiS1/6000 ／ Right-angled Connector

Cable Type : D, Connector Direction : B

Non-Flex /Flex	No. of axis	Standard /NRTL/CE	Length	Specification	Wire		
					Power	Brake	Pulsecoder
Flex	1	Standard /NRTL	7m	A05B-2607-J310	Diameter: Φ.4mm Weight: 0.05kg/m	Diameter: 6.4Φm Weight: 0.05kg/m	Diameter: Φ7mm Weight: 0.09kg/m
			14m	A05B-2607-J311			
			20m	A05B-2607-J312			
			30m	A05B-2607-J313			
		CE	7m	A05B-2607-J315	Diameter: Φ.4mm Weight: 0.07kg/m	Diameter: Φ.4mm Weight: 0.07kg/m	Diameter: Φ7mm Weight: 0.09kg/m
			14m	A05B-2607-J316			
			20m	A05B-2607-J317			
			30m	A05B-2607-J318			

Available Servo Motor: αiF1/5000, αiF2/5000, αiS2/5000, αiS4/5000, βiS2/4000, βiS4/4000

Cable Type : E, Connector Direction : E

Non-Flex /Flex	No. of axis	Standard /NRTL/CE	Length	Specification	Wire	
					Power・Brake	Pulsecoder
Flex	1	Standard /NRTL	7m	A05B-2607-J200	Diameter: Φ11.8mm Weight:0.2kg/m	Diameter: Φ7mm Weight:0.09kg/m
			14m	A05B-2607-J201		
			20m	A05B-2607-J202		
			30m	A05B-2607-J203		
		CE	7m	A05B-2607-J210	Diameter: Φ12.5mm Weight:0.25kg/m	Diameter: Φ7mm Weight:0.09kg/m
			14m	A05B-2607-J211		
			20m	A05B-2607-J212		
			30m	A05B-2607-J213		

Available Servo Motor: αiF4/5000, αiF8/3000, αiS8/4000, αiS12/4000, βiS8/3000, βiS12/3000

Cable Type : F, Connector Direction : D

Non-Flex /Flex	No. of axis	Standard /NRTL/CE	Length	Specification	Wire		
					Power	Brake	Pulsecoder
Flex	1	Standard /NRTL	7m	A05B-2607-J220	Diameter: Φ11.8mm Weight: 0.2kg/m	Diameter: 6.4Φmm Weight: 0.05kg/m	Diameter: Φ7mm Weight: 0.09kg/m
			14m	A05B-2607-J221			
			20m	A05B-2607-J222			
			30m	A05B-2607-J223			
		CE	7m	A05B-2607-J230	Diameter: Φ12.5mm Weight: 0.25kg/m	Diameter: Φ6.4mm Weight: 0.07kg/m	Diameter: Φ7mm Weight: 0.09kg/m
			14m	A05B-2607-J231			
			20m	A05B-2607-J232			
			30m	A05B-2607-J233			

（Note） In Cable of type F, select the motor with Brake.

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Available Servo Motor: αiF12/4000, αiF22/3000, βiS22/2000

Cable Type : F, Connector Direction : D

Non-Flex /Flex	No. of axis	Standard /NRTL /CE	Length	Specification	Wire		
					Power	Brake	Pulsecoder
Flex	1	Standard /NRTL	7m	A05B-2607-J240	Diameter: Φ2.9mm Weight: 0.31kg/m	Diameter: Φ.4mm Weight: 0.05kg/m	Diameter: Φ7mm Weight: 0.09kg/m
			14m	A05B-2607-J241			
			20m	A05B-2607-J242			
			30m	A05B-2607-J243			
		CE	7m	A05B-2607-J250	Diameter: Φ2.9mm Weight: 0.31kg/m	Diameter: Φ.4mm Weight: 0.07kg/m	Diameter: Φ7mm Weight: 0.09kg/m
			14m	A05B-2607-J251			
			20m	A05B-2607-J252			
			30m	A05B-2607-J253			

（Note） In Cable of type F, select the motor with Brake.

Available Servo Motor: αiF30/4000, αiF40/3000, αiS22/4000, αiS30/4000, αiS40/4000

Cable Type : F, Connector Direction : D

Non-Flex /Flex	No. of axis	Standard /NRTL/CE	Length	Specification	Wire		
					Power	Brake	Pulsecoder
Flex	1	Standard /NRTL	7m	A05B-2607-J260	Diameter: Φ20mm Weight: 0.61kg/m	Diameter: Φ6.4mm Weight: 0.05kg/m	Diameter: Φ7mm Weight: 0.09kg/m
			14m	A05B-2607-J261			
			20m	A05B-2607-J262			
			30m	A05B-2607-J263			
		CE	7m	A05B-2607-J270	Diameter: Φ20mm Weight: 0.64kg/m	Diameter: Φ6.4mm Weight: 0.07kg/m	Diameter: Φ7mm Weight: 0.09kg/m
			14m	A05B-2607-J271			
			20m	A05B-2607-J272			
			30m	A05B-2607-J273			

（Note） In Cable of type F, select the motor with Brake.

8.6.FANUC Servo Torch Cable

Cable Type : G

Servo Torch	Interface	Non-Flex /Flex	Standard /CE	Length	Specification	Wire	
						Power	Pulsecoder
For aluminum Standard spec. A05B-1220-J301	W/F1	Non-Flex	Standard	7m	A05B-2607-J350	Diameter: Φ6.4mm Weight: 0.05kg/m	Diameter: Φ7mm Weight: 0.09kg/m
				14m	A05B-2607-J351		
For soft steel Standard spec. A05B-1220-J302				20m	A05B-2607-J352		
				30m	A05B-2607-J353		
For aluminum CE spec. A05B-1220-J303	Motor Connection	Non-Flex	CE	Inquire of the Robot laboratory.			
For soft steel CE spec. A05B-1220-J304							

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8.7.Servo hand cable (HAND P/HAND M cable、ARP/ARM cable)

Available Servo Motor: αiF1/5000, αiF2/5000, αiS2/5000, αiS4/5000,
βiS0.4/5000, βiS0.5/6000, βiS1/6000

Cable Type: I, Interface: D

Non-Flex / Flex	No. of axis	Standard /NRTL /CE	Length	Specification	Wire	
					Power/Brake (ARM)	Pulsecoder (ARP)
Non-Flex	3	Standard /NRTL	7m	A05B-2632-J320	Diameter: Φ15.2mm(#C) Weight : 0.35kg/m	Diameter: Φ14.2mm(#C) Weight: 0.31kg/m
			14m	A05B-2632-J321		
			20m	A05B-2632-J322		
			30m	A05B-2632-J323		

Cable Type: I, Interface: D

Non-Flex / Flex	No. of axis	Standard /NRTL /CE	Length	Specification	Wire	
					Power/Brake (ARM)	Pulsecoder (ARP)
Non-Flex	6	Standard /NRTL	7m	A05B-2632-J350	Diameter: Φ15.2mm(#C) Weight : 0.35kg/m	Diameter: Φ14.2mm(#C) Weight: 0.31kg/m
			14m	A05B-2632-J351		
			20m	A05B-2632-J352		
			30m	A05B-2632-J353		

Cable Type: I, Interface: D

Non-Flex / Flex	No. of axis	Standard /NRTL /CE	Length	Specification	Wire	
					Power/Brake (ARM)	Pulsecoder (ARP)
Non-Flex	9	Standard /NRTL	7m	A05B-2632-J380	Diameter: Φ15.2mm(#C) Weight : 0.35kg/m	Diameter: Φ14.2mm(#C) Weight: 0.31kg/m
			14m	A05B-2632-J381		
			20m	A05B-2632-J382		
			30m	A05B-2632-J383		

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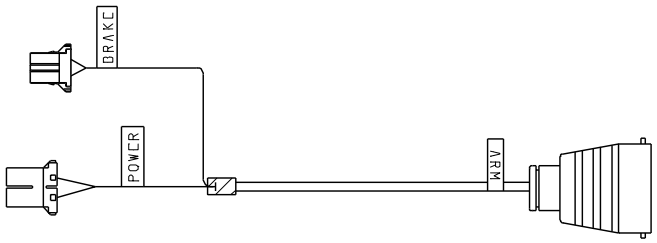
8.8. Cable Type

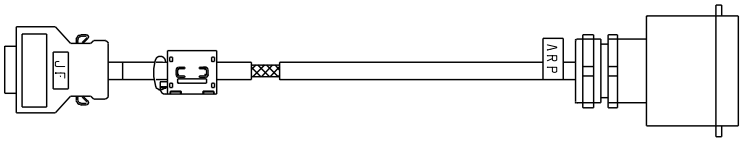
The outline shape of each Cable Type is shown as follows.

＜Cable Type A＞

① Power/Brake Cable

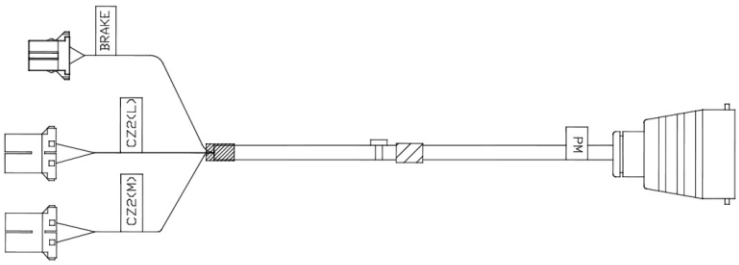


② Pulsecoder Cable

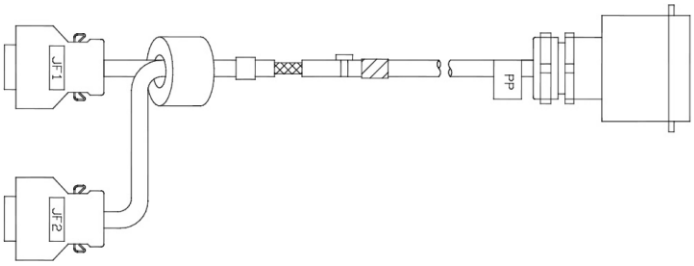


＜Cable Type B＞

① Power/Brake Cable

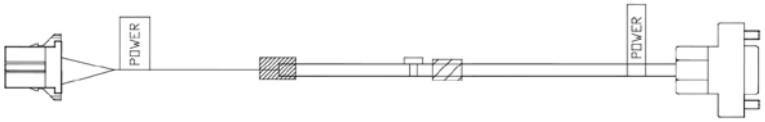


② Pulsecoder Cable

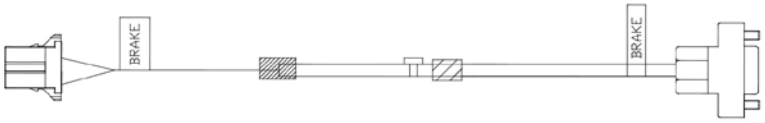


＜Cable Type C＞

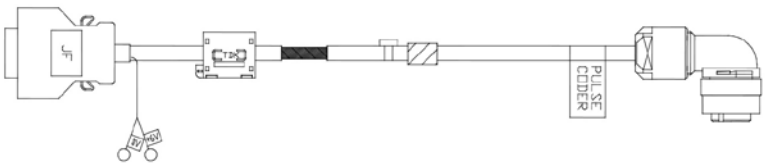
① Power



② Brake Cable



③ Pulsecoder Cable



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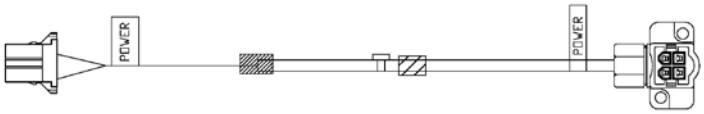
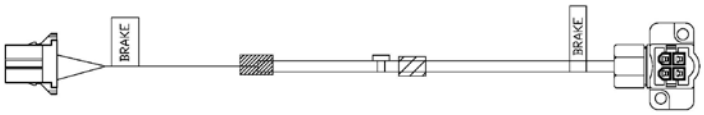
＜Cable Type D＞

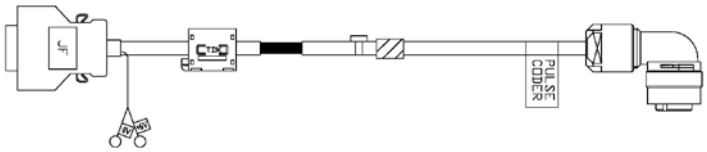
① Power



② Brake Cable

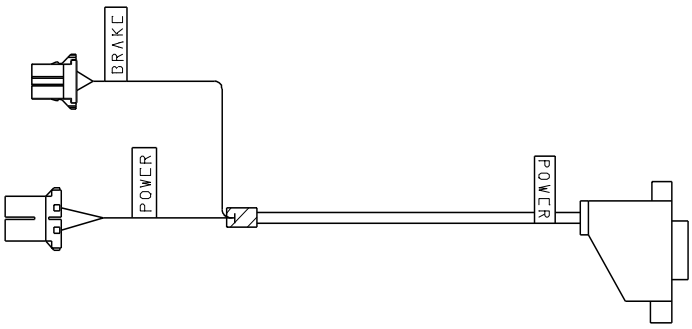


③ Pulsecoder Cable

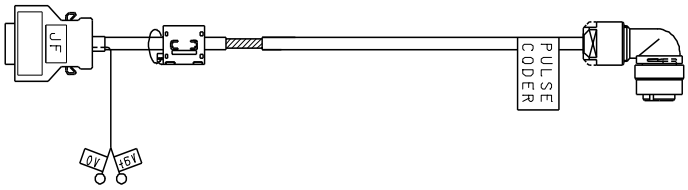


＜Cable Type E＞

① Power/Brake Cable

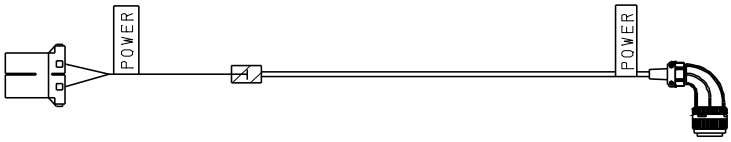


② Pulsecoder Cable

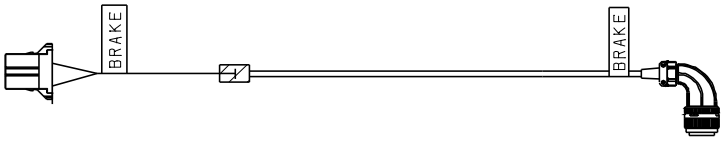


＜Cable Type F＞

① Power Cable



② Brake Cable



③ Pulsecoder Cable

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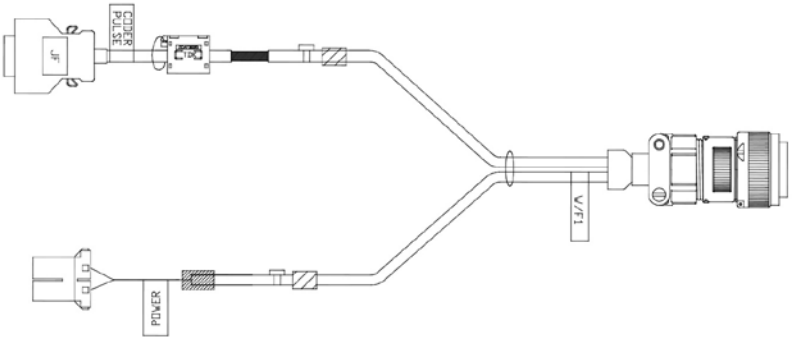
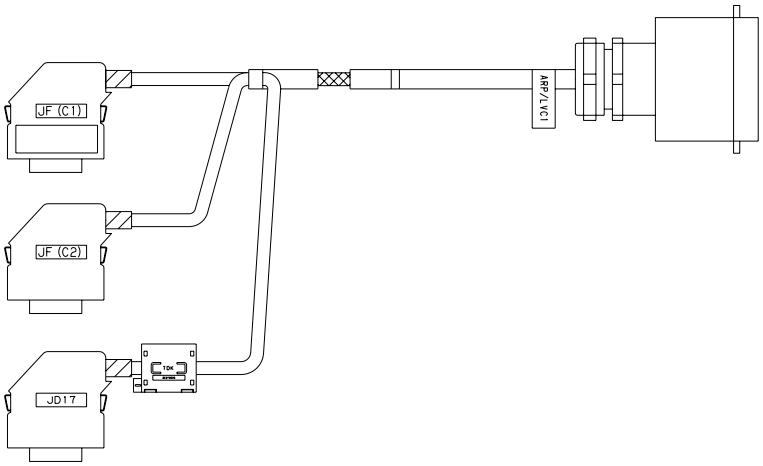
＜Cable Type G＞

① Power/Pulsecoder Cable

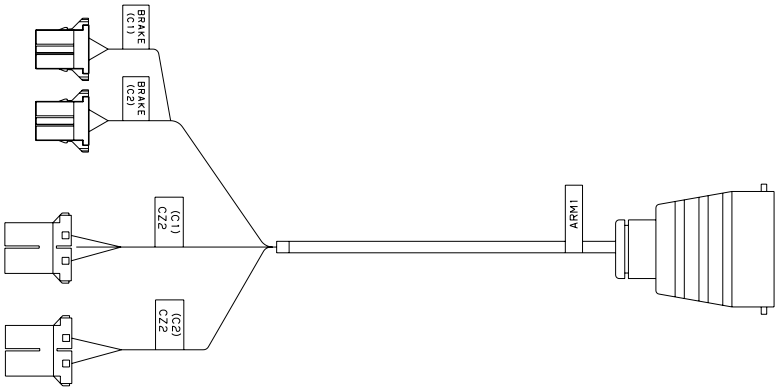


＜ Cable Type H＞

① Power/Brake Cable



② Pulsecoder Cable



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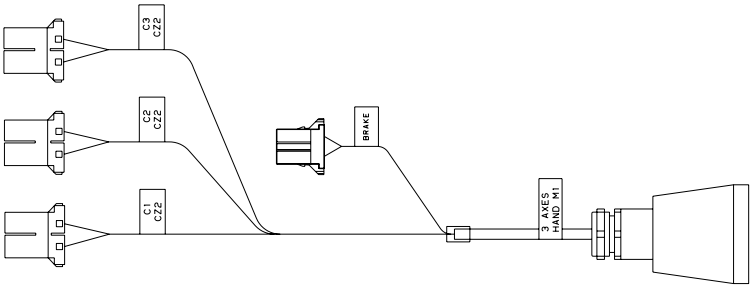
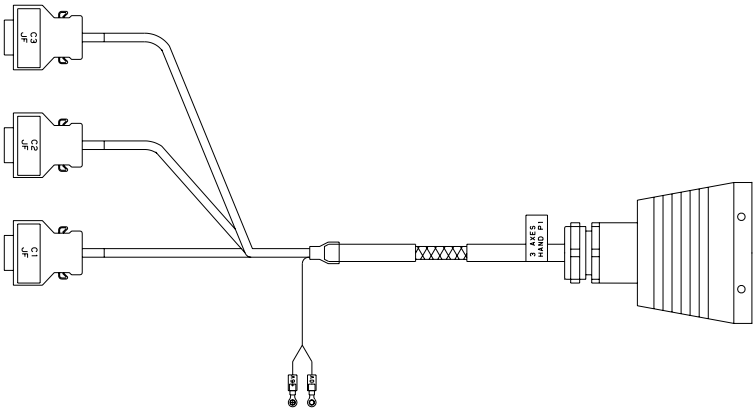
＜ Cable Type I＞

① Power/Brake Cable



② Pulsecoder Cable



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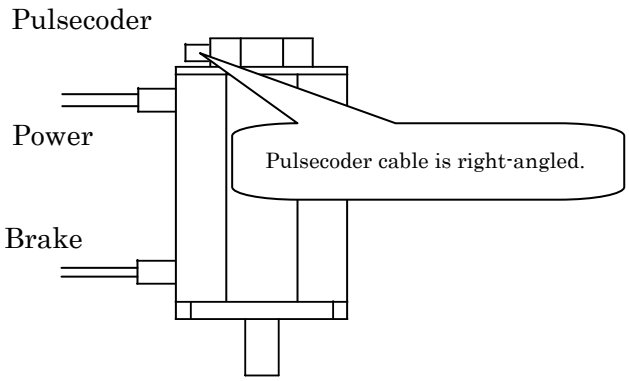
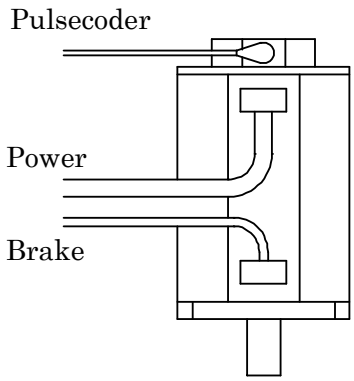
8.9. Connector Direction of Servo Motor

Connector directions of servo motors are as follows. This drawings shows servo motor direct connection cable.

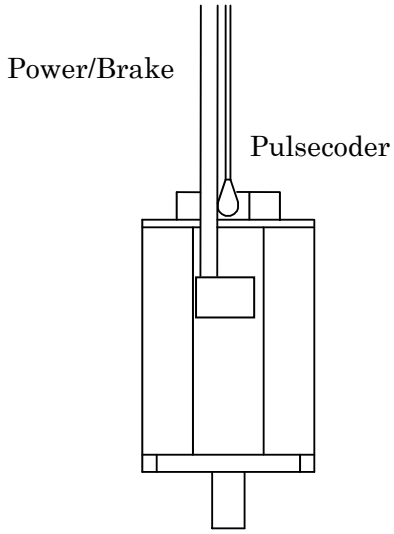
＜Connector Type A＞



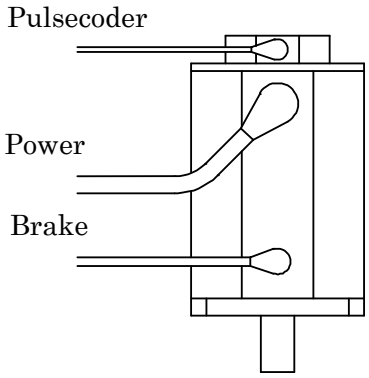
＜Connector Type B＞



＜Connector Type E＞



＜Connector Type D＞



8.10. Auxiliary axis Interface

Interfaces of each Auxiliary axis are shown below.

Followings figure is shown the connector view from wire side.

●shows the code pin position.

＜Interface Type A＞

① ARP1 (Han24DD)

				G	SHIELD						
1	(SPD1)	5	0V	9	+5V	13	(SPD2)	17	0V	21	+5V
2	(*SPD1)	6	0V	10	+5V	14	(*SPD2)	18	0V	22	+5V
3	RD1	7		11		15	RD2	19		23	
4	*RD1	8	(0VA)	12	(6VA)	16	*RD2	20	(0VA)	24	(6VA)

● ●

(Note) The hatching shows the signal of the second axis. Therefore, they are blanks in one axis specification.

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② ARM1 (Han15D)

A1	U1	B1	BKP1	C1	U2
A2	V1	B2	BKM1	C2	V2
A3	W1	B3	BKP2	C3	W2
A4	G1	B4	BKM2	C4	G2
A5		B5		C5	
●		G	(SHIELD)		●

(Note) The hatching shows the signal of the second axis. Therefore, they are blanks in one axis specification.

＜Interface Type B＞

① PP (Han24DD)

				G	SHIELD						●
1	(SPD1)	5	0V	9	+5V	13	(SPD2)	17	0V	21	+5V
2	(*SPD1)	6	0V	10	+5V	14	(*SPD2)	18	0V	22	+5V
3	RD1	7		11		15	RD2	19		23	
4	*RD1	8	(0VA)	12	(6VA)	16	*RD2	20	(0VA)	24	(6VA)
●											

② PM (Han K12/2, 12HsC)

		G	(SHIELD)		
2	U2	1	U1	3	BKP1
5	V2	4	V1	6	BKM1
8	W2	7	W1	9	BKP2
11	G2	10	G1	12	BKM2

＜Interface Type C＞

① J7P (Han24DD)

●				G	SHIELD						
1	(SPD1)	5	0V	9	+5V	13		17		21	
2	(*SPD1)	6	0V	10	+5V	14		18		22	
3	RD1	7		11		15		19		23	
4	*RD1	8		12		16		20		24	
●											

② J7M (Han K12/2, 12HsC)

		G	(SHIELD)		
2	U1	1	U1	3	BKP1
5	V1	4	V1	6	BKM1
8	W1	7	W1	9	
11	G1	10	G1	12	

(Note) The hatching shows the signal of large motor. Therefore, they are blanks in small motor specification.

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＜Interface Type D＞

Apply to servo hand interface (1Axis-9Axes)
Note）HAND P/HAND M (ARP/ARP) is prevented incorrect connection by guide pin or guide bush.

Configuration of Servo hand interface

Servo hand 1 to 3 Axes：HAND P1 / HAND M1
Servo hand 4 to 6 Axes：HAND P1 ,P3 / HAND M1 , M3
Servo hand 7 to 9 Axes：HAND P1 , P3 , P5 / HAND M1 , M3 , M5

①HAND M1(ARM1)：(Han24DD)：FEMALE／1 Lever (Robot connection cable side)

				G	SHIELD						
1	BKP1	5	BKM1	9	U1	13	V1	17	W1	21	G1
2	BKP2	6	BKM2	10	U2	14	V2	18	W2	22	G2
3	BKP3	7	BKM3	11	U3	15	V3	19	W3	23	G3
4		8		12		16		20		24	

②HAND M3(ARM3)：(Han24DD)：FEMALE／1 Lever (Robot connection cable side)

				G	SHIELD						
1	BKP4	5	BKM4	9	U4	13	V4	17	W4	21	G4
2	BKP5	6	BKM5	10	U5	14	V5	18	W5	22	G5
3	BKP6	7	BKM6	11	U6	15	V6	19	W6	23	G6
4		8		12		16		20		24	

③HAND M5(ARM5)：(Han24DD)：FEMALE／1 Lever (Robot connection cable side)

				G	SHIELD						
1	BKP7	5	BKM7	9	U7	13	V7	17	W7	21	G7
2	BKP8	6	BKM8	10	U8	14	V8	18	W8	22	G8
3	BKP9	7	BKM9	11	U9	15	V9	19	W9	23	G9
4		8		12		16		20		24	

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④HAND P1(ARP1):(Han42DD):FEMALE／2 Lever (Robot connection cable side)

				G	SHIELD						
1	SPD1	8	SPD2	15	SPD3	22		29	0V1	36	0V3
2	*SPD1	9	*SPD2	16	*SPD3	23		30	0V1	37	0V3
3	SPRQ1	10	SPRQ2	17	SPRQ3	24		31	0V2	38	
4	*SPRQ1	11	*SPRQ2	18	*SPRQ3	25		32	0V2	39	
5	5V1	12	5V2	19	5V3	26		33		40	
6	5V1	13	5V2	20	5V3	27		34	(6V)	41	(6V)
7	(0V)	14	(0V)	21	(0V)	28		35	(6V)	42	

⑤HAND P3(ARP3):(Han42DD):FEMALE／2 Lever (Robot connection cable side)

				G	SHIELD						
1	SPD4	8	SPD5	15	SPD6	22		29	0V4	36	0V6
2	*SPD4	9	*SPD5	16	*SPD6	23		30	0V4	37	0V6
3	SPRQ4	10	SPRQ5	17	SPRQ6	24		31	0V5	38	
4	*SPRQ4	11	*SPRQ5	18	*SPRQ6	25		32	0V5	39	
5	5V4	12	5V5	19	5V6	26		33		40	
6	5V4	13	5V5	20	5V6	27		34	(6V)	41	(6V)
7	(0V)	14	(0V)	21	(0V)	28		35	(6V)	42	

⑥HAND P5(ARP5):(Han42DD):FEMALE／2 Lever (Robot connection cable side)

				G	SHIELD						
1	SPD7	8	SPD8	15	SPD9	22		29	0V7	36	0V9
2	*SPD7	9	*SPD8	16	*SPD9	23		30	0V7	37	0V9
3	SPRQ7	10	SPRQ8	17	SPRQ9	24		31	0V8	38	
4	*SPRQ7	11	*SPRQ8	18	*SPRQ9	25		32	0V8	39	
5	5V7	12	5V8	19	5V9	26		33		40	
6	5V7	13	5V8	20	5V9	27		34	(6V)	41	(6V)
7	(0V)	14	(0V)	21	(0V)	28		35	(6V)	42	

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8.9. Flexible cable condition

8.9.1 Minimum bend radius of cable

The minimum bend radius of Flex Cable is 200(mm).

8.9.2. Flex Cable

Flex cable must be used with cable carrier for moving part. The problem such as shortening of the longevity of Cable if Cable carrier is not used occurs.

8.9.3. Cable carrier

- The radius (R) of cable carrier should be more than 200mm.
- The cable should be Non-Flex to cable carrier by using the clamp. (e.g. Rubber packing)
- The size of hole to support cable inside cable carrier should be more than 110% of cable size and should have more than 3mm gap.
- The cable should be mounted inside cable carrier without twisting.

8.9.4. Cable Durability Test

The result of cable durability test is indicated as follows.

[Test method : U form bending test]

- Connect all wires and shield in series, and measure the resistance of wire and shield.
- If resistance of wire and shield is 20% larger than initial resistance, the test is regarded as failure.

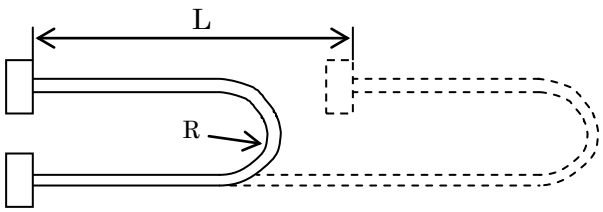

Fig Cable durability test

8.9.5. Experimental Result

Experimental results are as follows.

Experimental results

Material No.	Point	Result after 500,000 times	
		Resistance up (%)	Sample dismantlement result
No.1	Wire	0	There is no disconnection
	Shield	0	
No.2	Wire	0	There is no disconnection

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8.10.6 Wring in the cable carrier

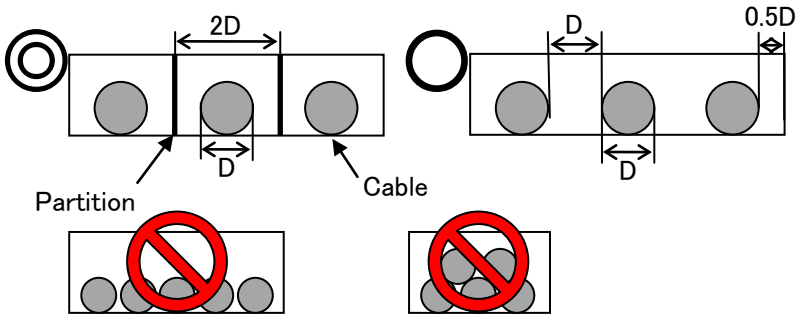
NOTE:
- If the precautions described in this section are not fulfilled, meandering or disconnection of cable may occur.
- Confirm the usage and notifications of cable carrier.
- If cutting fluid(oil) or chips are possible to adhered to the cable, use the fully enclosed cable carrier.
- Confirm that the cable is installed without twisting after installation and operation.

8.10.6.1 Partition of cable carrier

Separate each cable by a partition in the cable carrier.

Do not cross the cable or overlap each other.

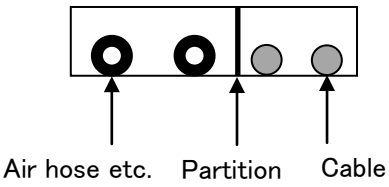
Select the cable carrier which has enough space and keep cable occupancy to 30% or less. If partition is used, keep the interval 2D or more. And if partition is not used, keep the clearance of each cables D or more. (D is the diameter of cable).



NOTE: In case of below, use the partition.

- In case of mixing with the air hose etc.

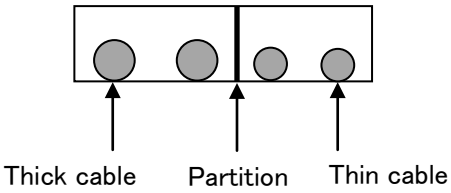
Be sure to separate the cable and the air hose etc. with the partition when the oil pressure and air hose etc. is wired to the cable carrier.



- In case of mixing with the cable of much different thickness

Since the thick cable affect the thin cable, the thin cable may be occurred the issue.

Be sure to separate the thin cable and the thick cable with the partition when the cable of much different thickness is wired to the cable carrier.



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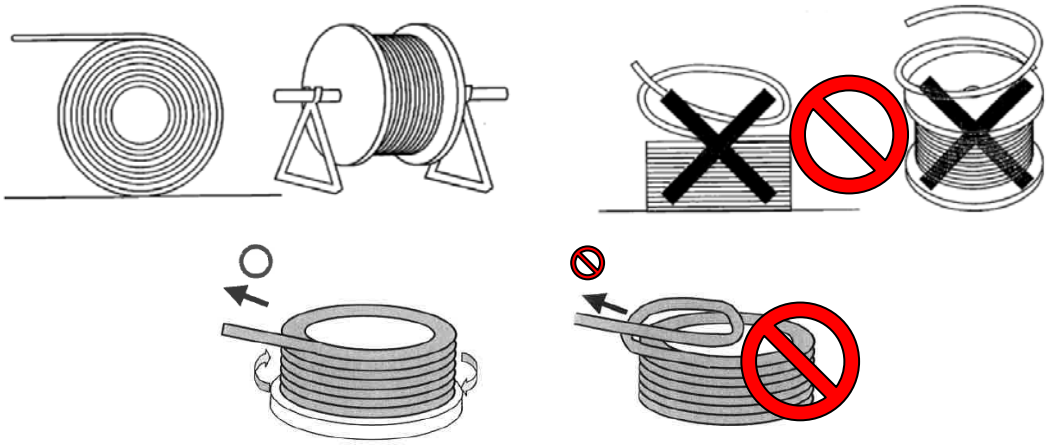
8.10.6.2 Install procedure of cable to the cable carrier

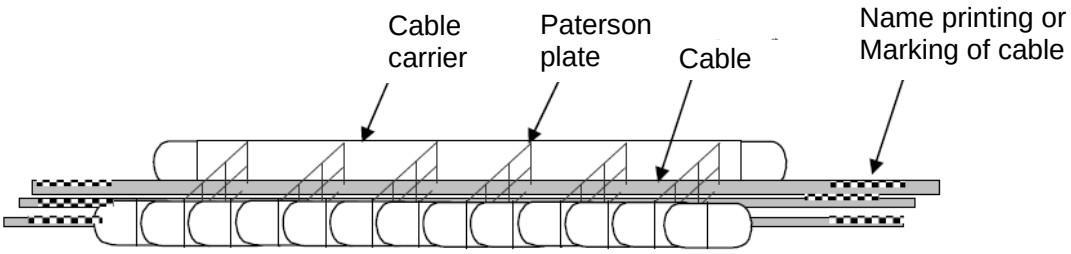
1. Extract the cable with the turntable etc. Confirm that the cable is not twisted and put straight without twisting.

NOTE: Extracting the cable with the turntable etc. as below drawing can avoid twisting of cable. Extracting straight the cable without the turntable etc. cause meandering or decreasing the bending characteristics because twisting remain in the cable.

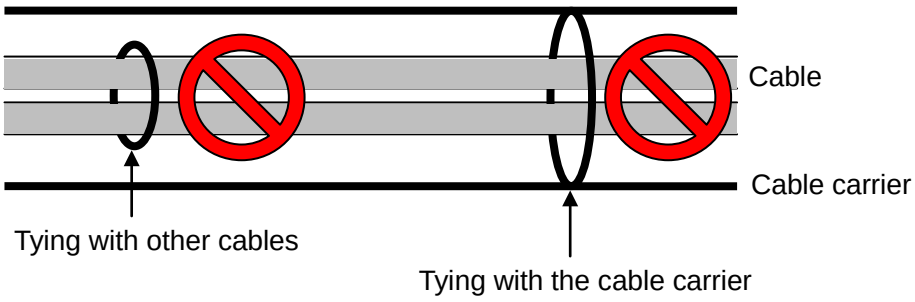


Correct extraction method　　wrong extraction method

2. Install the cable into the cable carrier without twisting.
3. Install the cable according to name printing or marking on the cable sheath, and confirm that the cable is installed without twisting after installation and operation.



NOTE:　Do not tie with other cables in the cable carrier.



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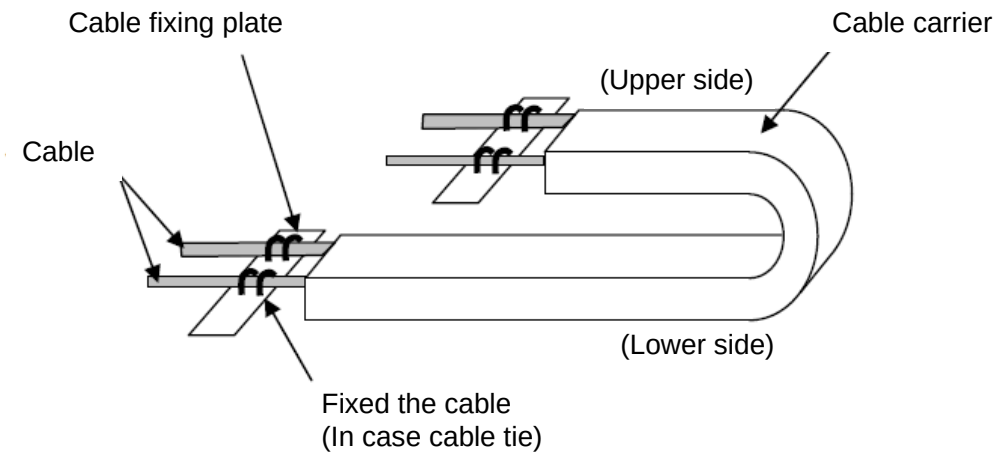
8.10.6.3 Fixing the cable

After installing the cable to cable carrier, fix the cable by the following procedure.

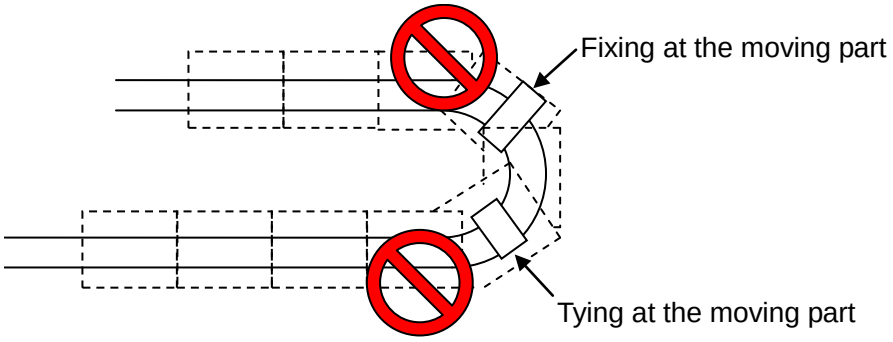
1. Fixing the cable at the upper side

   Fix the cable individually at the upper termination side as below drawing. Do not fix the cable at moving part.

   If cable tie is used to fix the cable, recommend to use two set of cable tie.



If the cable is fixed at the moving part of cable carrier, the cable life is decreased by stress to the cable.



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2. Adjusting the cable length

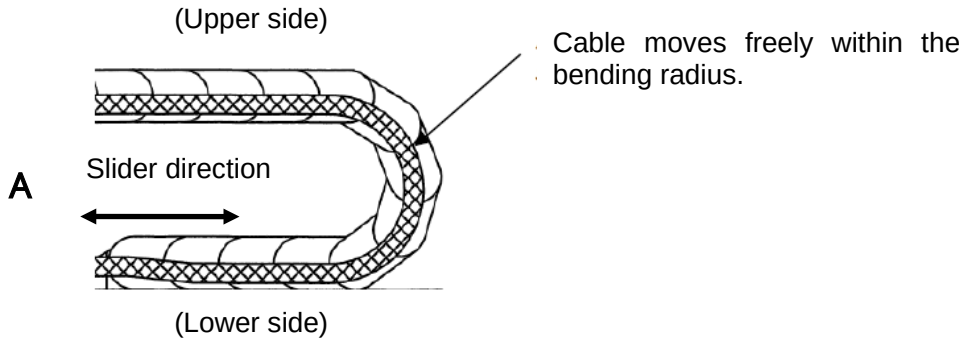
Adjust that the cable is installed with some looseness of the slider direction.

The cable should not contact to the inner surface of the cable carrier at the moving part tightly (Below drawing A).

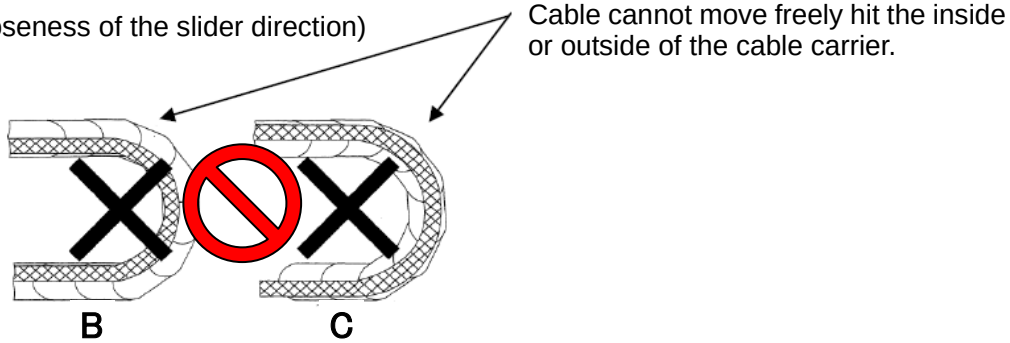
It becomes easier to adjust to the state of the below drawing A by the adjusting the position of the lower side cable and the adjusting the position of the cable to the middle of the below drawing B and C.

If the cable is too short as the below drawing B or too long as the below drawing C, the cable sheath is damaged because the cable is rubbed the cable carrier and the cable is contacted with the other cables.

Good example (with looseness of the slider direction)



Bad example
(without looseness of the slider direction)



3. Fixing the cable at the lower side

Fix the cable individually at the lower termination side.

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9. Battery Unit

Select Battery Unit for Auxiliary axis.

However, it is unnecessary only for ARP/ARM Cable, J7P/J7M Cable, and PP/PM Cable.

Battery Unit	Specification
Battery box with size D batteries (x4)	A05B-2607-J460

10. Gun change / Brake

10.1 Gun change / Brake Unit

R-30iB/R-30iB Plus controller has the brake circuit on 6-channel amplifier for 2 auxiliary axis brakes (Brake Group 1). Brake Group #1 is the brake that is worked with robot simultaneously. If robot controller needs more brake circuit or the brake circuit that its group is not Brake Group #1 (that isn't worked with robot simultaneously), specify the brake unit based on the followings.

(1) 6-channel amplifier : Brake group #1×2 axes

(2) Brake / Gun change unit : Brake group #2×1 axis / Brake group #3×2 axes (Note 1)

(3) 2 group 4 Axes Brake unit : Brake group #2×2axes / Brake group #3×2 axes

(4) 2 group 8 Axes Brake unit : Brake group #2×2axes / Brake group #3×4 axes / Brake group #4×2 axes

(5) 3 group 8 Axes Brake unit : Brake group #2×2 axes / Brake group #3×4 axes / Brake group #4×2 axes

(6) Hand change unit (1 Axis) : Brake group #2×1 axis / Brake group #3×2 axes / Brake group #4×2 axes/Maximum available brake are 4 axes.

(7) Hand change unit (2 Axes) : Brake group #2×2axes / Brake group #3×2 axes / Brake group #4×2 axes/Maximum available brake are 4 axes.

(8) Hand change unit (3 Axes) : Brake group #2×3axes / Brake group #3×2 axes / Brake group #4×2 axes/Maximum available brake are 4 axes.

Brake number (connector)	Specification	#1 (CRR65AB) Standard	#2 (CRR65C) Option	#3 (CRR65D) Option	#4 (CRR65E) Option	Location	Limitation
6 channel amplifier (1)	–	X2	–	–	–	–	Correspond to all motors.
Brake / Gun change unit (2)	A05B-2607-J400	X2	X1 (Note 3) (for Gun change) (Note 5)	X2 (Note 2)	–	A	Note2,Note3
2 group 4 Axes Brake unit (3)	A05B-2607-J410	X2	X2	X2	–	A	Correspond to all motors.
2 group 8 Axes Brake unit (4)	A05B-2607-J412	X2	X4	X4	–	A	Correspond to all motors.
3 group 8 Axes Brake unit (5)	A05B-2607-J416	X2	X2	X4	X2	A	Correspond to all motors.
Hand change unit (1 Axis) (6)	A05B-2607-J402	X2	X1 (Note 4) (for Gun change)	X2	X2	A	
Hand change unit (2 Axes) (7)	A05B-2607-J408+ A05B-2603-J035(Plate)	X2	X2 (Note 4) (for Gun change)	X2	X2	A+B	
Hand change unit (3 Axes) (8)	A05B-2607-J409+ A05B-2603-J035(Plate)	X2	X3 (Note 4) (for Gun change)	X2	X2	A+B	

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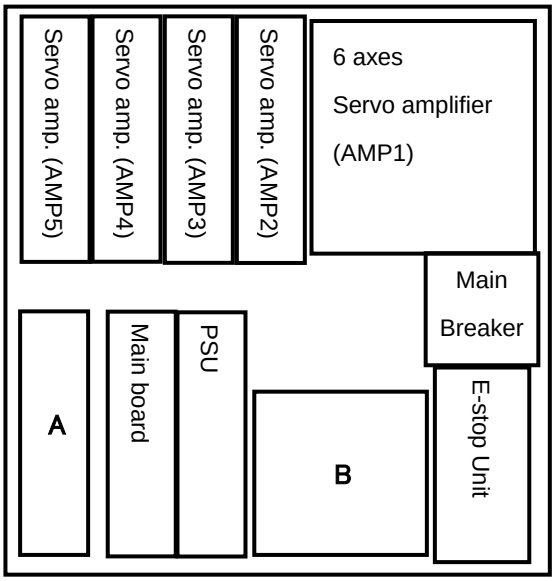
Location of brake unit



(Note 1) The brake units((2)-(8)) can be used with the brake circuit of 6 channel amplifier(1).

(2) set of the brake units((2)-(8)) cannot be used simultaneously.

(Note 2) Brake / Gun change unit cannot be applied to small size Motor.

Refer to the following available motors for Brake / Gun change unit.

αiF 4/5000, αiF 8/3000, αiF 12/4000, αiF 22/3000, αi F 30/4000, αi F 40/3000,
αiS 8/4000, αiS 12/4000, αiS 22/4000, αi S 30/4000, αiS 40/4000,
βiS 8/3000, βiS 12/3000, βiS 22/2000

In case of the brake for small motors, use the brake circuit of 6 channel amplifier(1) or brake unit(3).

(Note 3) Gun change axis cannot be applied to small size Motor.

Refer to the following for available motors for Gun change axis.

αiS 8/4000

Contact the design section about other motors with gun change axis.

(Note 4) Hand change unit cannot be applied to small size Motor.

Refer to the following for available motors for Hand change unit.

αiF 1/5000, αiF 2/5000, αiS 2/5000, αiS 4/5000, βiS 0.4/5000, βiS 0.5/6000
βiS 1/6000, βiS 2/4000, βiS 4/4000, βiS 8/3000

Contact the design section about other motors with hand change unit.

(Note 5) Even if Gun change is not used, the brake #2 can be used.

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10.2. Gun change cable

Select the cable for Gun change.

Cable	Specification
Gun change cable	A05B-2607-J401

(Note) Gun change cable is not needed for Hand change (A05B-2607-J402, A05B-2607-J408, A05B-2605-J409).

10.3. Spare Fuse

Specify each preliminary fuse when Brake / Gun change unit or Brake Unit is specified.

Fuse	Specification
For Brake Unit or Brake / Gun change unit (2A)	A05B-2605-K002

11. Transformer upgrade

Refer when you upgrade the capacity of Transformer from Standard by "2.3. Limitation of servo amplifier and configuration".

It is necessary to change Transformer, Breaker, and E-stop Unit for upgrade of Transformer capacity.

When AC Reactor is used, the upgrade is unnecessary.

(Note 1) When Aux. Axis and Robot or several Aux. Axis do not move together, no need to apply upgrade of Transformer.

(Note 2) When Robot motion is very tight, LV or Excessive error may occur. Please select transformer with margin.

11.1 Transformer capacity

	3kVA ➡	7.5kVA ➡	10.5kVA ➡	13kVA
Type D Transformer (200-230V, 380-400V)	A05B-2603-H354	A05B-2603-H355	A05B-2603-H356	A05B-2603-H357
Type E Transformer (380-415, 440-480, 500-575V)	A05B-2603-H350	A05B-2603-H351	A05B-2603-H352	A05B-2603-H353

(Note) In case of type E transformer using with CE controller, primary power voltage should be 500V and under.

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11.2 Capacity of breaker

	3kVA ➡	7.5kVA ➡	10.5kVA ➡	13kVA
Breaker 200-230V	A05B-2603-H075	A05B-2603-H072	A05B-2603-H070	A05B-2603-H071
Breaker 380-400V	A05B-2603-H079	A05B-2603-H077	A05B-2603-H074	A05B-2603-H074
Breaker 380-500V	A05B-2603-H078	A05B-2603-H076	A05B-2603-H073	A05B-2603-H073
Leakage Breaker 200-230V	A05B-2603-H105	A05B-2603-H102	A05B-2603-H100	A05B-2603-H101
Leakage Breaker 380-400V	A05B-2603-H109	A05B-2603-H107	A05B-2603-H104	A05B-2603-H104
Leakage Breaker 380-500V	A05B-2603-H108	A05B-2603-H106	A05B-2603-H103	A05B-2603-H103
UL Breaker 200-230V	A05B-2603-H263	A05B-2603-H262	A05B-2603-H261	-
UL Breaker 380-575V	A05B-2603-H266	A05B-2603-H265	A05B-2603-H264	A05B-2603-H264
UL/CSA Breaker 380-480V	A05B-2603-H255	A05B-2603-H255	A05B-2603-H254	A05B-2603-H254
Breaker 200-230V（CE）	A05B-2603-H136	A05B-2603-H135	-	-
Breaker 380-500V（CE）	A05B-2603-H132	A05B-2603-H131	A05B-2603-H130	A05B-2603-H130
Leakage Breaker 200-230V（CE）	A05B-2603-H166	A05B-2603-H165	-	-
Leakage Breaker 380-500V（CE）	A05B-2603-H162	A05B-2603-H161	A05B-2603-H160	A05B-2603-H160

(Contact to design section in case of capacity upgrade is necessary for the controller with Integrated ROBOWELD controller)

11.3 E-stop Unit

	In case of transformer upgrade, 3kVA->7.5kVA, 3kVA->10.5kVA, 3kVA->13kVA	In case of transformer upgrade, 7.5kVA->10.5kVA, 7.5kVA->13kVA, 10.5kVA->13kVA
Standard	Change A05B-2603-H280 to A05B-2603-H282	Not required to upgrade.
UL / CSA	Change A05B-2603-H290 to A05B-2603-H292	

11.4 Regenerative resistor

In case of transformer upgrade, 3kVA->10.5kVA, 3kVA->13kVA, 7.5kVA->10.5kVA, 7.5kVA->13kVA	In case of transformer upgrade, 3kVA->7.5kVA, 10.5kVA->13kVA, In case of Power regeneration
Change A05B-2603-H060 to A05B-2603-H062	Not required to upgrade.

(Note）The upgrade is unnecessary for M-420iA.

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11.5 Upgrade of transformer capacity for A05B-2603-H053, -H054 (6 channel amplifier).

In case of the upgrade the capacity of controller with A05B-2603-H053 (ARC Mate 100iC, M-10iA etc.), A05B-2603-H054 (M-430iA etc.), specify the order according to the following notifications.

- The 2000W regenerative resistor cannot connect to the A05B-2603-H053, -H054 because this 6 channel amplifier has only (1) connector for 1000W regenerative resistor.

Transformer upgrade for ARC Mate 100iC, M-10iA, M-430iA etc.

Transformer capacity	Discharge resister 1000W	Discharge resister 2000W
3KVA	A05B-2603-H060	Not available
7.5KVA	A05B-2603-H060	Not available
10.5KVA	A05B-2603-H060	Not available
13KVA	A05B-2603-H060	Not available

11.6 Add Aux. Axis of the controller with AC reactor

If aux. axis is needed with AC reactor; confirm that SUMC does not exceed the limitation for AC reactor described in the previous chapter. If the required capacity calculated from SUMP and SUMA exceed the capacity of E-stop unit, then specify the E-stop unit according to the following table.

The capacity calculated from SUMP and SUMA	3KVA	7.5KVA	10.5KVA	13KVA
AC reactor	available	available	available	Not available
E-stop unit (Standard)	A05B-2603-H280	A05B-2603-H282	A05B-2603-H282	Not available
E-stop unit (UL/CSA)	A05B-2603-H290	A05B-2603-H292	A05B-2603-H292	Not available

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12. Assembly specification

12.1 Amplifier assembly specification

SM- Code1 / Code2 / Code3

(1) Code1：Kind of specification

A：The servo amplifier assembly is shown.

(2) Code2：Servo amplifier number

A2：AMP2, A3：AMP3, A4：AMP4, A5：AMP5

(3) Code3：Servo amplifier specify

Specify H*** of servo amplifier specification A05B-2605-J040#H***(A06B-6240-H***).

Specification example 1

SM-A/A2/H103

To use A05B-2605-J040#H103(A06B-6240-H103) in AMP2 is show.

Specification example 2

SM-A/A3/H205

To use A05B-2605-J040#H205(A06B-6240-H205) in AMP3 is show.

12.2 PS for power regeneration assembly specification

SM- Code1 / Code2 / Code3

(1) Code1：Kind of specification

PS：PS assembly is shown.

(2) Code2：Servo amplifier number

A2：AMP2 (Whenever PS is selected, it is assembled to AMP2.)

(3) Code3：PS for power regeneration specify

Specify H*** of PS specification A05B-2605-J030#H***(A06B-6200-H***).

Specification example

SM-PS/A2/H015

To use A05B-2605-J030#H015(A06B-6200-H015) in AMP2 is show.

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12.3 Assembly instruction for Cables

SM- Code1 / Code 2 / Code 3 / Code 4

(1) Code1：Kind of specification

C：The cable assemblies other than the Gun change axis cable or PP/PM cable are shown.

GC1：The assembly of the Gun change axis cable is shown. (In case of using the one cable, the last number “1” of the code1 can be either attached or not attached.)

C1：The assembly of the J1 axis cable of PP/PM cable is shown.

C2：The assembly of the J1 axis cable of PP/PM cable is shown.

NC ： Non connection

(2) Code2：specification of cable

Underlined text of cable order specification, ”A05B-2607-J***（except “-“） is shown.

NONE：It is shown that the cable is not ordered.

(3) Code3：Connection

In case of Gun change axis, appoint to servo amplifier for Gun axis.

A2L：The cable is connected to L-axis of AMP2. In case the single-axis amplifier for AMP2 is used, A2L must be specified.

A2M：The cable is connected to M-axis of AMP2.

A2N：The cable is connected to N-axis of AMP2

A3L：The cable is connected to L-axis of AMP3. In case the single-axis amplifier for AMP3 is used, A3L must be specified.

A3M：The cable is connected to M-axis of AMP3.

A3N：The cable is connected to N-axis of AMP3

A4L：The cable is connected to L-axis of AMP4. In case the single-axis amplifier for AMP4 is used, A4L must be specified.

A4M：The cable is connected to M-axis of AMP4.

A4N：The cable is connected to N-axis of AMP4

A5L：The cable is connected to L-axis of AMP5. In case the single-axis amplifier for AMP5 is used, A5L must be specified.

A5M：The cable is connected to M-axis of AMP5.

A5N：The cable is connected to N-axis of AMP5

(4) Code4：brake group

65AB：Brake group 1 (same with brake of robot)

65C：Brake group 2

65D：Brake group 3

(NB or blank)：No brake cable is included or brake cable is not assembled.

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Specification example 1

SM-C/07J200/A2L/65AB

Signal and power cables of “A05B-2607-J200” are connected to L-axis of AMP2. Brake cable is connected to 6-channel servo amplifier.

Specification example 2

SM-C/07J180/A2M/65AB Rail unit cable

SM-C/07J200/A2L/65AB Cable for other axes

This system has two (2) auxiliary axes. Rail unit axis is assigned M-axis of dual axis servo amplifier.

Specification example 3

SM-C1/07J160/A2L/65AB

SM-C2/07J161/A2M/65AB

Signal and power cables of PP/PM (J1) are connected to L-axis of AMP2. Signal and power cables of PP/PM (J2) are connected to M-axis of AMP2. Brake cables are connected to 6-channel servo amplifier.

Specification example 4

SM-C1/07J160/A2L/65AB

SM-C2/07J161/A2M/65AB

Signal and power cables of PP/PM (J1) are connected to L-axis of AMP2. Signal and power cables of PP/PM (J2) are connected to M-axis of AMP2. Brake cables are connected to 6-channel servo amplifier.

Apply to servo hand cable (ARP/ARM cable)

In case of 4 axes servo hand cable (ARP/ARM cable) and R-2000iB/210F (Hand1-3: brake group2, Hand4: brake group3).

SM-C1/32J35*/A3L/65C	SM-C2/32J35*/A3M/65C	SM-C3/32J35*/A3N/65C
SM-C4/32J35*/A4L/65D	SM-C5/32J35*/NC	SM-C6/32J35*/NC

Specification example 5

SM-GC1/NONE/A2L/65C

This example is shown that the cable for Gun change is not ordered. This specification is used when the cable for Gun change is procured by customer.

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13. Specially auxiliary axis

The ordering list is different of the kind of Positioner.

13.1. FANUC 1 axis Positioner

Common Specification of FANUC 1 axis Positioner

Item	Name	Specification
Servo motor	-	unnecessary
Servo amplifier	αiSV80	A05B-2605-J040#H105
Auxiliary axis control	Auxiliary axis amplifier 1 axis	A05B-2607-J001
Adaptor plate	60mm(x1)	A05B-2607-J430
Servo card	12 axes	A05B-2600-H041 (for R-30iB)
		A05B-2670-H041 (for R-30iB Plus)
Brake Cable Tag	AMP1, CRR65AB	A05B-2605-J060
Motor Cable Tag	AMP2/L	A05B-2605-J070
Auxiliary axis Cable	-	Refer to following table
Battery unit	-	Refer to following table
Transformer	-	capacity upgrade is unnecessary
Resistance regeneration	-	capacity upgrade is unnecessary
Breaker	-	capacity upgrade is unnecessary
E-stop Unit	-	capacity upgrade is unnecessary
Amplifier assembly specification	-	SM-A/A2/H105
Cable assembly specification	-	SM-C/07J***/A2L/65AB (Note)

(Note) Match 07J*** to the specification of selected Cable.

Interface for Positioner

In Positioner, there are two kinds of Interface.

(Motor connecting directly Type and HARTING Connector relay Type)

Confirm the combination of Positioner, Cable, and the battery by the following table.

Regarding Interface, Cable and Battery for Positioner

1 axis Positioner	Positioner option	Cable specification for Auxiliary axis				Battery
		Interface	Motor	Non-Flex/Flex	Standard/CE	
1000kg/1500kg (Hollow Type) A05B-1220-J102	unnecessary	Motor direct	aiS22/4000	Flex	Selection	A05B-2507-J460
	A05B-1220-J114	J7P/J7M	aiS22/4000	Non-Flex・Flex	Selection	unnecessary
300kg/500kg (Hollow Type) A05B-1220-J103	unnecessary	Motor direct	aiS8/4000	Flex	Selection	A05B-2507-J460
	A05B-1220-J112	ARP/ARM	aiS8/4000	Non-Flex・Flex	Selection	unnecessary
1000kg/1500kg (Compact Type) A05B-1220-J104	unnecessary	Motor direct	aiS22/4000	Flex	Selection	A05B-2507-J460
	A05B-1220-J116	J7P/J7M	aiS22/4000	Non-Flex・Flex	Selection	unnecessary

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13.2. FANUC 2 axes Positioner

2 axes servo Positioner for ARC WELDING(A05B-1220-J201)

Item	Name	Specification
Servo motor	-	unnecessary (Including Positioner)
Servo amplifier	αiSV80/80	A05B-2605-J040#H209
Auxiliary axis control	Auxiliary axis amplifier for first axis	A05B-2607-J001
Adaptor plate	60mm(x1)	A05B-2607-J430
Servo card	12 axes	A05B-2600-H041 (for R-30iB)
		A05B-2670-H041 (for R-30iB Plus)
Brake Cable Tag	AMP1, CRR65AB	A05B-2605-J060
Motor Cable Tag	AMP2/L	A05B-2605-J070
Motor Cable Tag	AMP2/M	A05B-2605-J071
Auxiliary axis Cable	PM/PP Cable	Refer to “8.3. PM/PP Cable“
Battery Unit	-	Unnecessary
Transformer	-	capacity upgrade is unnecessary
Resistance regeneration	-	capacity upgrade is unnecessary
Breaker	-	capacity upgrade is unnecessary
E-stop Unit	-	capacity upgrade is unnecessary
Amplifier assembly specification	-	SM-A/A2/H209
Cable assembly specification	-	SM-C1/07J***/A2L/65AB (Note)
Cable assembly specification	-	SM-C2/07J***/A2M/65AB (Note)

(Note) Match 07J*** to the specification of selected Cable.

Match the amplifier number to the assembly position of amplifier.

Match the Brake number to connected Brake Connector.

13.3. FANUC Servo Torch

Item	Name	Specification
Servo motor	-	unnecessary (Including Servo torch)
Servo amplifier	αiSV20	A05B-2605-J040#H103
Auxiliary axis control	Auxiliary axis amplifier for first axis	A05B-2607-J001
Adaptor plate	60mm(x1)	A05B-2607-J430
Adaptor plate	60mm w/o/ Fin	A05B-2605-J420
Servo card	12 axes	A05B-2600-H041 (for R-30iB)
		A05B-2670-H041 (for R-30iB Plus)
Brake Cable Tag	-	Unnecessary
Motor Cable Tag	AMP2/L	A05B-2605-J070
Auxiliary axis Cable	Servo torch Cable	Refer to “8.6 FANUC Servo Torch Cable”
Battery Unit	-	Unnecessary
Transformer	-	capacity upgrade is unnecessary
Resistance regeneration	-	capacity upgrade is unnecessary
Breaker	-	capacity upgrade is unnecessary
E-stop Unit	-	capacity upgrade is unnecessary
Cable assembly specification	-	SM-A/A2/H103
Cable assembly specification	-	SM-C/07J35*/A2L (Note)

(Note) Match 07J*** to the specification of selected Cable.

Match the amplifier number to the assembly position of amplifier.

Match the Brake number to connected Brake Connector.

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<Appendix> R-30*i*B/R-30*i*B Plus Auxiliary Axis Config. Sheet

Customer :
Destination :
E No. :

Requester :
Date :
Robot :

Type : Standard, NRTL, CE Cabinet : A, B
Transformer: 3k, 7.5k, 10.5kVA, AC Reactor
Power regeneration, Resistance regeneration
Shipping : Factory , Retrofit

Axis No.	Servo Amp		AMP No.	Current (A)	Motor	Cable		I/F	Axis Control	Group No.	Brake Group	Purpose	Remarks
1	6 axes servo amplifier		AMP1	–	–	–		RP1,RM1	12-axes card	G1-J1	1	Robot	
2										G1-J2			
3										G1-J3			
4										G1-J4			
5										G1-J5			
6										G1-J6			
7													
8													
9													
10													
11													
12													
13													
14													
15													
16													

(Note) This table is needed when auxiliary axis set up. FANUC recommend this table is attached to robot.

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<Appendix> R-30*i*B/R-30*i*B Plus CONTROLLER VARIATION TABLE

Refer when the transformer upgrade
and "Resistor discharge" -> " Power regeneration", etc.

Name	Specification	R-2000iB	R-2000iC	R-1000iA	M-710iC	M-420iA	ARC Mate 100iC ARC Mate 100iD	ARC Mate 120iC	ARC Mate 120iD	M-10iA M-10iD	M-20iA M-20iB	R-2000iB/100P R-2000iC/100P		M-410iB M-410iC	R-2000iB/200T/220U R-2000iC/210L/270F	R-1000iA/120F-7B R-1000iA/130F	M-900iA/350 M-900iB/360	M-900iA/150P		M-900iA/600 M-900iB/700	M-2iA	M-3iA	M-430iA/4FH	M-430iA/2P/2PH	F-200iB	CR-35iA
Cabinet	-	A/B	A/B	A/B	A/B	A/B	A/B	A/B	A/B	A/B	A/B	A	B	A/B	A/B	A/B	A/B	A	B	B	A	A	A/B	A/B	A/B	A
Power transformer 3kVA	-						O	O	O	O	O												O	O		O
Power transformer 7.5kVA	-	O	O	O	O	O															O	O			O	
Power transformer 10.5kVA	-											O	O	O	O	O	O	O	O							
Power transformer 13kVA	-																			O						
AC Reactor (200-230VAC)	A05B-2601/2603-H365	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O		O	O	O	O	O	O
6 axes Servo Amplifier A	A05B-2601/2603-H050													O	O		O	O	O	O						
6 axes Servo Amplifier B	A05B-2601/2603-H051	O		O	O	O		O			O	O	O			O						O				O
6 axes Servo Amplifier C	A05B-2601/2603-H052		O						O												O					
6 axes Servo Amplifier D	A05B-2601/2603-H053						O			O															O	
6 axes Servo Amplifier E	A05B-2601/2603-H054																						O	O		
Servo Amplifier for Robot axis (Alpha iSV80/80, J4/J5)	A05B-2605-J040#H209																			O						
Servo Amplifier for Robot axis (Alpha iSV80, J6)	A05B-2605-J040#H105																			O						
Servo Amplifier for Robot axis (Alpha iSV4/4, J4/J5)	A05B-2605-J040#H201																						O			
Servo Amplifier for Robot axis (Alpha iSV4/4/4, J4/J5/J6)	A05B-2605-J040#H301																							O		
Resistor discharge A (6.5ohm/1 element)	A05B-2601/2603-H060	O	O	O	O		O	O	O	O	O										O	O	O	O	O	O
Resistor discharge B (6.5ohm/2 elements)	A05B-2601/2603-H062					O							O	O	O	O	O		O	O						
Power regeneration (Alpha i PS15)	A05B-2605-J030#H015	O	O	O	O	O						O	O	O	O	O	O	O	O		O	O				
Power regeneration (Alpha iPS37)	A05B-2605-J030#H037																			O						

(Note) Either "Power transformer" or "AC Reactor" is specified.
Either "Resistor discharge" or " Power regeneration" is specified.

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